

Anchoring Large Language Models to Empirical Behavioral Evidence

A Hybrid Approach for DCE-Grounded Narrative Policy Simulation

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Abstract

Objective: To develop and validate empirical-frontier regularization (EFR), an inference-time procedure anchoring LLM outputs to a DCE-estimated reference surface without retraining.

Methods: A Wuhan vaccination DCE ($N = 1,027$) was combined with LLM simulation on a 64-state policy grid. EFR computes $\pi_{\text{EFR}} = \lambda \bar{\pi}_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ with $\lambda = 0.25$ fixed as a conservative operating point.

Results: Held-out validation on six DCE choice tasks ($N = 205$ respondents) shows multinomial log loss: Pure-DCE 0.911, EFR 0.923, Raw LLM 1.020. EFR reduced log loss by 9.5% versus Raw LLM and approached Pure-DCE performance (within 1.3%). The unconstrained LLM violated waiting-time monotonicity on 62.5% (Qwen2.5-72B), 34.4% (DeepSeek V4 Pro), and 10.4% (MiroThinker) of scenarios; after EFR, MVR-Wait dropped to 30.2%, 7.3%, and 8.3% respectively. Rank correlation ρ rose from 0.061/0.302/0.531 (Raw) to 0.689/0.812/0.756 (EFR), all exceeding the uniform-noise null ($\rho = 0.367$, 95% CI [0.145, 0.563]); two of three models also exceeded the stricter permutation null that preserves the LLM’s marginal distribution (Supplement B).

Limitations: Single-site DCE; respondent-level holdout evaluates generalization to unseen respondents, not unseen tasks. Out-of-design scenarios lacked human ground truth.

Conclusion: EFR constrains LLM outputs to a DCE-derived frontier, bounding deviation while preserving narrative-extension capability. It is a wrapper for existing DCEs, not a replacement.

Keywords: empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment; behavioral simulation

Highlights

1. EFR is an inference-time regularization layer that constrains LLM outputs to a DCE-estimated reference without retraining.
2. On held-out respondent validation, EFR recovers most of the DCE model's predictive performance while retaining a bounded LLM component.
3. EFR improves structural consistency (monotonicity, rank order) but does not establish behavioral validity for out-of-design narrative scenarios.

Article Type: Methodological Article

1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for health-care policy^{19,18,27}. Their appeal is that they can interpret rich natural-language scenarios that conventional structured decision models cannot. However, linguistic fluency does not guarantee behavioral reliability: an unconstrained LLM may assign higher acceptance to a longer-waiting policy than to an otherwise identical shorter-waiting policy, contradicting empirically observed preference patterns and producing unreliable policy rankings.

Current approaches do not explicitly enforce DCE-estimated behavioral constraints^{1,12,20}: LLM agents may produce decisions that appear linguistically plausible while violating basic preference relationships implied by observed human behavior.

One source of such constraints is the discrete choice experiment (DCE), which estimates how observed human preferences vary with decision attributes⁵. In this study, we treat DCEs not as models to be replaced by LLMs, but as sources of DCE-estimated behavioral reference. A DCE-estimated choice frontier defines the preference ordering and directional relationships supported by observed responses. The challenge is to combine this empirical behavioral knowledge with the representational flexibility of LLMs, so that simulated predictions can incorporate an LLM-derived narrative component while remaining closer to DCE-implied structure; whether the resulting narrative extension is behaviorally more accurate on out-of-design scenarios is not established by the present study, because no human ground truth is available outside the original DCE design space.

We propose a **hybrid approach** that anchors LLM-generated choice probabilities to empirical behavioral evidence from DCEs. The method estimates an empirical choice frontier from DCE responses and blends the LLM’s raw probability with the frontier through a transparent weighted-averaging procedure. The adjustment is auditable: the raw LLM probability, the empirical frontier value, the blending weight, and the final adjusted probability are all directly inspectable. The method requires no retraining. The same LLM can be anchored using different empirical datasets, enabling domain-specific grounding while preserving general capabilities.

We evaluate the approach using a real-world vaccination-timing DCE involving 1,027 adults in

Wuhan, China. This study makes three contributions:

1. We introduce a lightweight inference-time adjustment procedure for grounding LLM-based simulation in empirically estimated human behavioral knowledge.
2. We show that the proposed adjustment improves structural consistency across multiple LLMs, reducing behavioral inconsistencies and improving agreement with the DCE-estimated reference. On held-out-respondent validation (local Ollama Qwen2.5-72B replication), EFR materially improved predictive performance relative to the unconstrained LLM (9.5% reduction in log loss) and approached the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component. **The retention of an LLM component is motivated by narrative-extension capability, not by evidence that LLMs provide accurate behavioral simulation for novel scenarios; the OOD evaluation demonstrates constraint effects, not validation of LLM behavioral accuracy.**
3. We demonstrate that empirically grounded behavioral models can be extended to richer natural-language healthcare policy scenarios while preserving auditable behavioral constraints.

2 Related Work

Our work connects two research areas: LLM-based synthetic agents for healthcare simulation and empirical behavioral models derived from discrete choice experiments (DCEs). While these two directions have developed largely independently, both seek to improve the realism and reliability of decision support. Our work lies at their intersection by incorporating empirically estimated behavioral knowledge into LLM-based synthetic agents through inference-time regularization.

Prior LLM-for-HTA work in this journal. Recent issues of *Value in Health* have featured LLMs along trajectories distinct from the behavioral-simulation axis we pursue: evidence extraction²², sponsor-HTA conversation simulation¹⁷, microsimulation²⁴, and LLM-based prediction of stated preferences⁶. We deploy LLMs as narrative-extension mechanisms anchored to a DCE-derived

empirical frontier, not as tools for review or negotiation. The closest analogue outside HTA is Twin-2K-500²⁵, which validates LLM-as-consumer simulation in marketing; we extend this line to health policy. A pre-existing LLM-powered social digital twin framework¹⁴ is conceptually adjacent but does not anchor to a pre-collected empirical choice frontier.

2.1 LLM-Based Synthetic Agents

Large language models (LLMs) have enabled a new generation of synthetic agents capable of generating natural-language reasoning and simulating human decision making across diverse domains^{3,19,18}. In healthcare, these agents have been explored for public health communication, policy evaluation, and behavioral simulation^{21,7}. Their principal advantage is representational flexibility: unlike conventional statistical models, LLMs can process rich textual descriptions and respond to scenarios that extend beyond predefined structured variables.

However, behavioral realism remains a major challenge: LLM responses may violate expected directional relationships (e.g., waiting-time monotonicity, where longer waits should reduce acceptance) or produce implausible preference orderings^{1,12,11}. Prompt engineering can reduce some inconsistencies but provides no explicit mechanism for enforcing behaviorally grounded constraints^{29,28}; the mechanism cannot guarantee compliance across diverse narrative phrasings outside the original prompt template. We adopt the LLM-as-narrative-component framing of Argyle et al.², Horton¹²: the LLM’s narrative output is one input to a convex blend with an empirical choice frontier, not a stand-alone behavioral model.

2.2 Empirical Behavioral Models from Discrete Choice Experiments

Discrete choice experiments (DCEs) provide a well-established framework for estimating how individuals trade off policy-relevant attributes such as waiting time, vaccine efficacy, side effects, and financial incentives^{23,5}. Under the random utility framework^{26,16}, DCE responses can be used to estimate an empirical choice frontier that preserves interpretable relationships between decision attributes and observed choices. Mixed-logit models further account for preference heterogeneity

across respondents^{9,10}.

In this study, we treat the DCE frontier as an empirically estimated source of behavioral knowledge rather than as assumption-free ground truth. While empirical choice models provide transparent and behaviorally interpretable predictions within their original experimental design, they cannot directly operate on rich natural-language policy scenarios.

2.3 Bridging Empirical Behavioral Models and LLM-Based Agents

Existing LLM-based synthetic agents provide representational flexibility but lack mechanisms for enforcing empirically estimated behavioral constraints^{15,24}. DCE-derived behavioral models provide transparent and interpretable structure but remain confined to their experimental design. Existing approaches trade flexibility for reliability or vice versa.

3 Methods

This study integrated two methodological components: (1) a discrete choice experiment to estimate empirical vaccination preferences, and (2) a procedure for aligning LLM predictions with these empirical patterns. The following sections describe each component and the validation approach.

3.1 Overview

Section 3.2 describes the DCE design and empirical frontier; Section 3.3 presents the EFR framework. Section 4.2 reports the 64-state evaluation, and Section 4.3 reports the out-of-design narrative evaluation; Section 5.6 concludes.

3.2 Stage 1: Empirical Behavioral Knowledge Extraction

The first stage extracts empirical behavioral knowledge from a vaccination-timing DCE conducted among 1,027 adults in Wuhan, China (fielded December 2022–January 2023). This knowledge is

an empirical choice frontier that serves as the behavioral reference for inference-time regularization.

DCE design and sampling. The Wuhan vaccination-timing DCE used a **non-orthogonal fractional design** with five policy-relevant attributes: waiting time (0, 1, 3, 6 months), vaccine efficacy (50%, 70%, 95%), side-effect risk (0.1%, 1%, 10%), cash incentive (50, 200, 800 RMB), and vaccine origin (Domestic, Imported). The intended non-opt-out candidate space contained 216 profile cells ($4 \times 3 \times 3 \times 3 \times 2$). All 1,027 respondents completed the **same six choice tasks**, each presenting two non-opt-out alternatives and an opt-out alternative; the design therefore contained six unique administered choice tasks covering 12 of the 216 candidate cells (5.6% coverage). Task presentation order was randomized across respondents to mitigate order effects. The design was generated using D-efficiency criteria with prior parameter estimates from pilot data; a post-hoc dominance audit confirmed that none of the six administered A-versus-B profile pairs exhibited strict dominance across the monotonic attributes.

Sampling and ethics. Respondents were recruited through an online panel provider (quota sampling: age 18–65, stratified by gender and education to approximate Wuhan adult population distribution). Eligibility criteria: adult resident of Wuhan, no prior COVID-19 vaccination. The full DCE questionnaire and recruitment protocol are available in the public repository. Individual-level DCE responses are not publicly released because of participant confidentiality. The study was reviewed and approved by the relevant institutional ethics committee, and all participants provided informed consent before participation.

Stakeholder involvement and instrument design. Two physician interviews and two rounds of twelve cognitive interviews (ISPOR good-research-practices^{13,4}) refined the choice-task framing.

Construction of the Empirical Behavioral Reference. The empirical behavioral reference was constructed from a 6-parameter grouped/conditional logit with five attribute coefficients and an opt-

out alternative-specific constant, estimated on the Wuhan DCE. The frontier maps each candidate policy to a DCE-implied binary policy-versus-opt-out probability (i.e., the probability of choosing the policy alternative over the opt-out under the conditional logit model), providing a deterministic signal that anchors the LLM’s narrative choices.

3.3 Stage 2: Inference-Time Adjustment Procedure

The adjustment algorithm aligns LLM outputs with the empirical reference by computing $\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g)$ (the EFR equation $\pi_{\text{adj}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$; full derivation in Appendix A.1 of the Supplementary Material). This binary policy-versus-opt-out formulation is the primary definition; for the exact-task categorical validation, EFR is extended to a convex blend of full three-alternative probability vectors (Appendix N of the Supplementary Material).

The final stage applies the adjusted probability to policy evaluation, comparing policy scenarios and estimating changes in expected choice probability.

For a baseline scenario $\mathbf{x}$ and an intervention that changes attribute x_k to x'_k , the policy response is:

$$\Delta_k(\mathbf{x}) = \pi_{\text{adj}}(y = 1 \mid \mathbf{x}') - \pi_{\text{adj}}(y = 1 \mid \mathbf{x}),$$

where $\mathbf{x}'$ denotes the post-intervention scenario. Averaging over comparable scenarios gives the mean policy response for attribute k .

Because evaluation operates only on the adjusted probability surface, the approach can be deployed locally without transmitting individual-level DCE responses to external providers. The resulting policy-response quantities are descriptive summaries and should not be interpreted as causal treatment effects.

Table 1: Behavioral consistency with the DCE-estimated reference before and after EFR on the 64-state grid. The reference surface is the DCE-estimated conditional-logit model evaluated on the grid attributes (wait $\in \{0, 2, 4, 6\}$, efficacy $\in \{0.3, 0.5, 0.7, 0.9\}$, side-effect level $\in \{0, 1, 2, 3\}$; cash and origin held at reference values). This is a **structural diagnostic** on a model-implied projection surface; the grid extends beyond the directly administered DCE profiles. Spearman’s ρ measures alignment with the DCE-estimated reference; the exact-task categorical validation (Figure 1) is the primary held-out benchmark.

Model	Method	MSE $\downarrow$	MAE $\downarrow$	MVR-Wait (%) $\downarrow$	Spearman’s ρ $\uparrow$
Qwen2.5-72B-Instruct	Raw LLM	0.0258	0.1269	62.5	+0.061
	EFR	0.0016	0.0317	30.2	0.689
DeepSeek V4 Pro	Raw LLM	0.0379	0.1570	34.4	0.302
	EFR	0.0024	0.0392	7.3	0.812
MiroThinker-1.7-mini	Raw LLM	0.0872	0.2644	10.4	0.531
	EFR	0.0054	0.0661	8.3	0.756

4 Experimental Evaluation

4.1 Experimental Setup

We evaluated EFR using three open-source LLMs (Qwen2.5-72B-Instruct, DeepSeek V4 Pro, MiroThinker-1.7-mini; endpoint and access details in Table S2). Decoding parameters were matched across models where the API permitted, with Qwen2.5-72B-Instruct and DeepSeek V4 Pro sampled at temperature = 0.7, top_p = 1.0, max_new_tokens = 512, and $R = 10$ repeated generations per state to average out sampling noise; MiroThinker-1.7-mini was sampled at temperature = 0.0 (deterministic), top_p = 1.0, max_new_tokens = 256, and $R = 10$ repetitions. The empirical choice frontier was estimated from a corrected 6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant on the full Wuhan DCE sample ($N = 1,027$ respondents) on vaccination timing (ISPOR good-research-practices^{13,8}). All LLM evaluations comply with the ViH/ISPOR ELEVATE-GenAI reporting framework (Appendix C).

4.2 Behavioral Consistency with the Empirical Frontier

Table 1 summarizes behavioral consistency on the 64-state evaluation grid. The grid attributes extend beyond the directly administered DCE profiles; the reference surface is a DCE-estimated conditional-logit projection. MVR is the wait-axis monotonicity violation rate; ρ is Spearman’s rank correlation between π and P_{emp} . The exact-task categorical validation (Figure 1) is the primary held-out benchmark. EFR ($\lambda = 0.25$) reduced MSE by 94%, MAE by 75%, and MVR-Wait substantially; all EFR predictions achieve $\rho \geq 0.69$, exceeding the uniform-noise null. The permutation null (Supplement B) distinguishes models whose LLM component shows detectable DCE-aligned rank information beyond mechanical anchoring (DeepSeek V4 Pro, MiroThinker; $p \leq .003$) from those that do not (Qwen2.5-72B; $p = .179$; evaluated on the local replication that reproduced the canonical $\rho = 0.689$). Alternative-level analysis: Table S7.

Table 2: Behavioral consistency under out-of-design policy scenarios.

Model	Method	MVR ($\downarrow$)	MAD ($\downarrow$)	Spearman ρ ($\uparrow$)
<i>Panel A: Core scenarios (6 paired policy descriptions)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.333 (2/6)	0.290	0.12
Qwen2.5-72B-Instruct	EFR	0.000	0.072	0.87
DeepSeek V4 Pro	Unconstrained LLM	0.667 (4/6)	0.257	-0.07
DeepSeek V4 Pro	EFR	0.000	0.064	0.87
<i>Panel B: Extended scenarios (14 paired; MVR-Wait defined only for 2 wait-varying pairs)</i>				
Qwen2.5-72B-Instruct	Unconstrained LLM	0.000 (0/2)	0.125	-0.01
Qwen2.5-72B-Instruct	EFR	0.000	0.031	0.93
DeepSeek V4 Pro	Unconstrained LLM	0.000 (0/2)	0.209	-0.06
DeepSeek V4 Pro	EFR	0.000	0.052	0.88

Note. Twenty paired scenarios (40 narrative descriptions total); eight vary waiting time and twelve vary only narrative content. MVR-Wait is reported on the eight wait-varying pairs (Panel A: 6; Panel B: 2); the twelve narrative-only pairs are reported as MAD/ ρ only. MAD is mean absolute deviation from P_{emp} . *Panel B Spearman ρ is reported descriptively only:* because P_{emp} contains only six distinct values across the 28 Panel B scenarios, ρ is strongly affected by ties and lies within the construction-only noise benchmark; it is therefore reported descriptively only.

4.3 Out-of-Design Policy Scenario Evaluation

We evaluated EFR on 20 narrative policy scenarios extending beyond the DCE design (Table 2). Panel A (six pairs) varies waiting time; Panel B (14 pairs) varies other attributes. Full scenario descriptions are provided in the Supplement.

Table 2 summarizes the out-of-design evaluation. In the core scenarios (Panel A), the unconstrained LLMs violated the expected waiting-time ordering in 2 of 6 paired scenarios for Qwen2.5-72B-Instruct and 4 of 6 paired scenarios for DeepSeek V4 Pro. After empirical-frontier regularization, no monotonicity violations were observed for either model, and rank agreement with the empirical behavioral reference increased substantially.

The differences became more pronounced in the extended scenarios (Panel B). Rich textual narratives containing fictional incentives and complex policy descriptions reduced the rank agreement of the unconstrained LLMs to approximately zero for both models. Among the two wait-varying pairs in Panel B, EFR maintained zero monotonicity violations (0/2); the twelve narrative-only pairs are reported as MAD and ρ only (all 20 paired scenarios = 40 narrative descriptions; only Qwen2.5-72B-Instruct and DeepSeek V4 Pro were evaluated on the OOD panel, MiroThinker-

1.7-mini was not). Rank agreement with the empirical reference was $\rho = 0.93$ for Qwen2.5-72B-Instruct and $\rho = 0.88$ for DeepSeek V4 Pro. These out-of-design results characterize structural consistency of the anchored outputs with the model-implied DCE reference surface; they do not establish predictive validity for human behavior outside the observed DCE design.

This experiment evaluates final anchored predictions; the separate contributions of language understanding and empirical regularization are not isolated here.

4.4 Policy Ranking Consistency

EFR increased agreement with the Pure-DCE policy ranking (a structural diagnostic, not an independent validation): Qwen2.5-72B-Instruct Spearman PRC $0.333 \rightarrow 0.867$; DeepSeek V4 Pro $0.133 \rightarrow 0.600$; MiroThinker-1.7-mini $0.533 \rightarrow 0.800$. Because the EFR blend contains 75% P_{emp} by construction, increased agreement with the Pure-DCE ranking is a mechanical consequence of the convex anchor, not an independent behavioral validation. PRC is therefore reported in the supplement as a structural diagnostic only; no significance test is performed on the 15 pairwise comparisons because they share six underlying policy effects and are not 15 independent Bernoulli trials.

5 Discussion

5.1 Principal Findings

EFR improves behavioral consistency without retraining. Three findings emerge. *First*, unconstrained LLMs produce choice probabilities poorly aligned with the DCE-estimated reference: waiting-time monotonicity violations ranged 10.4–62.5% (Qwen2.5-72B: 62.5%; DeepSeek V4 Pro: 34.4%; MiroThinker: 10.4%); rank correlation was weak (ρ : 0.061/0.302/0.531). *Second*, EFR reduced violations to 7.3–30.2% and raised ρ to ≥ 0.69 (0.689/0.812/0.756). Primary exact-task validation ($N = 205, 1,230$ choices) shows multinomial log loss: Pure-DCE 0.911, EFR

0.923, Raw LLM 1.020. EFR improved log loss versus Raw LLM (-0.0967 ; 95% CI -0.1468 to -0.0469); the EFR–Pure-DCE difference was small and imprecise ($+0.0121$; 95% CI -0.0234 to $+0.0472$). Alternative-level analysis ($N = 813$): Table S7. *Third*, policy-ranking agreement increased (PRC: $0.333 \rightarrow 0.867$, $0.133 \rightarrow 0.600$, $0.533 \rightarrow 0.800$); this is a structural diagnostic of the 75% P_{emp} blend, not independent validation (reported in supplement without significance testing).

5.2 Methodological Contributions

Our primary contribution is *inference-time empirical regularization*, a procedure that aligns LLM decisions with empirically estimated behavioral structure without modifying model weights. The design reflects theory-informed machine learning: established empirical models provide structural priors for generative AI systems. EFR is a wrapper, not a model: it can be applied to any LLM (closed-source or open-source) and any empirical choice frontier (mixed logit, latent class, or non-parametric). The convex combination $\pi_{\text{EFR}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ is the simplest possible anchor; future work could explore non-convex anchors (e.g., mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors).

A robustness stress test on eight Panel B scenarios across three smaller open-source models (4,800 prompts) corroborated this view: Qwen2.5-7B produced constant outputs (zero variance, default-answer signature), DeepSeek V4 (Flash variant) achieved 0% parse on the waiver-language N1 scenario. *These failures are a safety property, not evidence of LLM value-add*: when LLM output is degenerate, EFR continues to assign 75% of the weight to P_{emp} , bounding—but not eliminating—the LLM component (EFR is not a reliability gate, and the $\lambda = 0.25$ weight is fixed rather than adaptive). Our use of LLMs as anchored narrative components differs from prior tool-oriented LLM work: the empirical frontier, not the LLM, supplies the behavioral signal; the LLM component is permitted only in the convex blend with P_{emp} .

5.3 Implications for Health Policy Analysis

The EFR framework has three HTA implications. *First*, a recent DCE unlocks a low-marginal-cost narrative-policy simulation pipeline: HTA agencies need only the LLM API, DCE estimates, and a deterministic blend. This may be useful for exploratory screening when immediate collection of a new DCE is impractical. *Second*, EFR extends the productive life of an existing DCE: EFR applies to narrative policy descriptions (fictional incentives, side-effect profiles, communication strategies) outside the administered structured DCE profiles. *Third*, the framework is best suited to early-stage screening, not high-stakes final decisions; for the latter, EFR outputs should be triangulated with traditional stated-preference methods.

Limitations and design considerations. (i) *Wuhan DCE, single-jurisdiction sample.* The reference surface is estimated from one DCE fielded in a single Chinese city (collection period and full DCE questionnaire reported in the public code repository at <https://github.com/REDACTED/behavioral-digital-twins>); preference stability across pandemic phases is not established. Replication across additional jurisdictions and time periods is planned future work (Conclusion). (ii) *Conditional-logit specification and frontier stability.* The static frontier uses a 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample, not a random-coefficients mixed logit. Respondent-cluster bootstrap (2,000 replicates) confirms frontier stability: Spearman ρ between bootstrap and canonical frontier has mean 0.988 (median 0.993; 95% CI [0.948, 1.000]) and all six coefficients retain their sign in $\geq 97.7\%$ of replicates (Appendix J of the Supplementary Material). The corrected opt-out ASC narrows P_{emp} to [0.47, 0.64]; we trade within-DCE-design contrast for a behaviorally honest specification. (iii) $\lambda = 0.25$ as an illustrative conservative operating point. The blending weight is a fixed illustrative conservative operating point preserving a non-zero LLM contribution while keeping the empirical DCE dominant; it is not a test-set-optimized predictive optimum. The held-out DCE log loss and Brier score are reported as sensitivity diagnostics and were not used to select λ . Structurally, this weight retains only one quarter of the raw LLM–frontier deviation pointwise, me-

chanically bounding frontier-relative squared deviation to $\lambda^2 = 6.25\%$ of the unconstrained LLM value (Supplementary Material F.2). **The paper’s substantive conclusions derive from the full λ sensitivity analysis (Supplementary Material F.3), not from the specific value $\lambda = 0.25$.** (iv) *Inference-time regularization vs. retraining.* EFR is a convex combination of two streams, not a learned gate; non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors) remain future work.

5.4 Why Not Use the Discrete Choice Model Alone?

EFR trades exact-task predictive performance for narrative-policy evaluation capability (multinomial log loss: Pure-DCE 0.911; EFR 0.923; Raw LLM 1.020; uniform benchmark 1.099). **EFR materially improved exact-task predictive performance relative to the unconstrained LLM** (9.5% reduction in log loss) and approached the predictive performance of the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component for narrative extension. EFR does not outperform the DCE model within the original structured choice domain, but it substantially improves on the unconstrained LLM and recovers most of the DCE model’s predictive performance while preserving an LLM component for narrative extension. The NDS pilot ($n = 2$) is reported as illustrative mechanism evidence only: isotonic NDS $\rho = -0.943$ vs. EFR $\rho = +0.894$ for DeepSeek V4 Pro. **Figure 1** reports the exact-task categorical held-out validation results.

5.5 Implementation and Policy Recommendations

Significance. EFR is a constraint wrapper that applies an existing DCE-derived frontier to narrative-policy evaluation. The fixed $\lambda = 0.25$ anchor bounds, rather than eliminates, the deviation of LLM outputs from the DCE-derived frontier. The OOD narrative evaluation demonstrates that EFR preserves structural consistency with the DCE-estimated reference (monotonicity, rank order), but this is a constraint effect, not evidence that the LLM component provides accurate behavioral simulation for novel policy scenarios. EFR is best understood as a narrative-extension wrapper for an existing DCE, not as a replacement for the underlying DCE model or as validation of LLMs as



Figure 1: Independent local replication validation multinomial log loss on the six administered DCE choice tasks (Pure-DCE: 0.911; EFR: 0.923; Raw LLM: 1.020; uniform benchmark: 1.099; $N = 205$ held-out respondents, 1,230 choices; local Ollama Qwen2.5-72B). In respondent-level paired bootstrap analyses, EFR improved log loss relative to Raw LLM (difference -0.0967 ; 95% CI, -0.1468 to -0.0469), whereas the EFR–Pure-DCE difference was small and imprecisely estimated ($+0.0121$; 95% CI, -0.0234 to $+0.0472$). Because all respondents completed identical tasks, this validation evaluates generalization to unseen respondents rather than to unseen choice tasks or attribute profiles.

standalone behavioral simulators. Implementation requires no model retraining: an analyst estimates a 6-parameter grouped/conditional logit from an existing DCE, generates LLM predictions, and blends them through $\pi_{\text{adj}} = \lambda \pi_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ ($\lambda = 0.25$). Three policy implications follow. *With a DCE*, EFR provides a path from evidence to narrative policy simulation. *Without a DCE*, the framework is not a substitute for stated-preference data. *Best for early-stage screening*, not high-stakes final decisions.

5.6 Conclusion

EFR anchors LLM outputs to DCE estimates without retraining. Independent local replication validation on the six administered DCE choice tasks ($N = 205$ respondents, 1,230 choices; local Ollama Qwen2.5-72B) shows multinomial log loss: Pure-DCE (0.911), EFR (0.923), Raw LLM (1.020), Uniform (1.099). EFR materially improved exact-task predictive performance relative to the unconstrained LLM (9.5% reduction in log loss) and approached the predictive performance of the Pure-DCE model (within 1.3%), while retaining a non-zero LLM component for narrative extension. EFR does not outperform the DCE model within the original structured choice domain, but it substantially improves on the unconstrained LLM and recovers most of the DCE model’s predictive performance while preserving an LLM component for narrative extension. On the structured 64-state grid, MVR-Wait fell from 10.4–62.5% to 7.3–30.2% across three LLMs (Qwen2.5-72B-Instruct: 62.5% $\rightarrow$ 30.2%; DeepSeek V4 Pro: 34.4% $\rightarrow$ 7.3%; MiroThinker-1.7-mini: 10.4% $\rightarrow$ 8.3%) under EFR anchoring at $\lambda = 0.25$ (Table 1). *In the eight wait-varying OOD narrative pairs (40 narrative descriptions total, evaluated with Qwen2.5-72B-Instruct and DeepSeek V4 Pro only), EFR eliminated the observed waiting-time reversals on Panel A (Qwen: 2/6 to 0/6; DeepSeek: 4/6 to 0/6) and held the Panel B wait-varying pairs at 0/2 for both models; the twelve narrative-only pairs (varying side-effect, efficacy, or fictional narrative content while holding wait constant) are reported as MAD and Spearman ρ only.* EFR is best understood as a constrained narrative-extension mechanism for an existing DCE, with the LLM component permitted only inside the convex blend with P_{emp} . The empirical reference, not the LLM, supplies the behavioral signal; whether the narrative extension is behaviorally more accurate on out-of-design scenarios remains to be established by future work with human ground truth. Three directions are planned future work: (i) replication across additional diseases, jurisdictions, and policy domains; (ii) exploration of non-convex anchoring mechanisms (e.g., mixture-of-experts gating); (iii) quasi-experimental validation with human ground truth on out-of-design narrative scenarios.

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Supplementary Materials

The following supplementary materials accompany this manuscript and are available in the online appendix.

A. Tables

- **Table S1.** Observed EFR Spearman ρ versus uniform-noise and permutation nulls on the 64-state grid.
- **Table S2.** LLM decoding and endpoint parameters used in the main, supplementary, and stress-test evaluations.
- **Table S3.** *Administered* DCE attribute levels: waiting time $\{0, 1, 3, 6\}$ months (4 real design levels; the raw coded data contain a single 1-row outlier at wait = 2 months from Respondent 1 only, which was excluded from all analyses before sample-size calculations). The analytic sample contains 18,485 alt-rows (6,162 choice tasks $\times$ 3 alternatives minus 1 anomalous row). All 1,027 respondents completed the same six choice tasks; task presentation order was randomized across respondents to mitigate order effects. Each task presented two non-opt-out alternatives and an opt-out alternative. The held-out validation subsample ($N = 205$ respondents) contains 1,230 valid tasks (205×6 ; the anomalous row was from Respondent 1, who was not in the held-out subsample). The design therefore contained six unique administered choice tasks and 12 non-opt-out profiles per respondent (12 unique non-opt-out profiles in the design; $12,324 = 1,027 \times 12$ profile instances across all respondents). The 12 administered non-opt-out attribute combinations per respondent enumerate 12 of the $3 \times 3 \times 3 \times 2 \times 4 = 216$ non-opt-out candidate cells (5.6% per-respondent coverage).
- **Table S4.** Attribute coding dictionary: DCE questionnaire display, estimation coding, 64-state evaluation-grid coding, and coefficient unit definitions.
- **Table S5.** Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier. Standard errors from BFGS Hessian; bootstrap 95% confidence intervals for the 6 coefficients.

- **Table S6.** λ -sensitivity for frontier-alignment diagnostics on the 64-state grid. Pure-DCE baseline, unconstrained LLM, and EFR Anchor evaluated across $\lambda \in \{0.00, 0.10, 0.20, 0.25, 0.30, 0.40, 0.50\}$.
- **Table S7.** λ -sensitivity for held-out DCE predictive validation on the 813 *alternative-level* rows subset (from 205 held-out respondents). Alt-level binary log loss, Brier score, and AUC across λ .
- **Table S8.** Data-scaling ablation: EFR Anchor ($\lambda = 0.25$) vs. unconstrained LLM across training fractions (30%, 50%, 70%, 100%).
- **Table S9.** Parse success rates across models and experiment scales.
- **Table S10.** Runtime summary for full-scale unconstrained LLM generation.
- **Table S11.** Policy Ranking Consistency (PRC) relative to the Pure-DCE reference; descriptive structural diagnostic, no significance test.
- **Table S12.** Respondent-level paired cluster bootstrap for exact-task multinomial log loss (six canonical tasks; $N = 205$ respondents).
- **Table S13.** Bootstrap intervals for selected frontier-alignment diagnostics.
- **Table S14.** Attribute-level mapping between the administered DCE design and the 64-state LLM evaluation grid. The matched subset is restricted to cells where the DCE design level and the grid level coincide as a float value (no rescaling, no interpolation).
- **Table S15.** Covariate balance between matched and unmatched held-out subsets.
- **Table S16.** Primary exact-task multinomial held-out log loss on six canonical DCE tasks ($N = 205$ respondents, 1,230 choices). Pure-DCE, EFR, Raw LLM, and uniform baseline.

B. Figures

- **Figure S1.** Data-scaling ablation: EFR Anchor ($\lambda = 0.25$) evaluated across DCE training shares (30%, 50%, 70%, 100%); companion to the data-scaling narrative in Appendix G.

C. Replication Materials

Replication materials are available at <https://github.com/REDACTED/behavioral-digital-twins>, including: the questionnaire, code, de-identified aggregate/grid-level inputs, model outputs, and derived analysis datasets required to reproduce the reported LLM and frontier-alignment analyses. Individual-level DCE responses are not publicly released because of participant confidentiality.

Online Supplement: Anchoring Large Language Models to Empirical Behavioral Evidence: A Hybrid Approach for DCE-Grounded Narrative Policy Simulation

Blinded for peer review

This supplement contains technical appendix material referenced in the main manuscript: derivation details, sensitivity analyses, and replication code listings.

A Technical Details

Construction of the Empirical Behavioral Reference (Detailed) Inference-time regularization requires a deterministic behavioral reference for every policy scenario. For the main experiments, we therefore construct a $4 \times 4 \times 4 = 64$ -state evaluation grid by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing the cash incentive and vaccine origin at their reference levels. The behavioral reference is computed using the MLE coefficients of the corrected 6-parameter grouped/conditional logit (five attribute coefficients plus an opt-out alternative-specific constant) on the full DCE sample; the canonical utility specification used throughout this paper is reported in Section 3.2 of the main manuscript and includes all five DCE attributes (waiting time, vaccine efficacy, side-effect risk, cash incentive, vaccine origin) plus the opt-out ASC:

$$U(x, \bar{\beta}) = \bar{\beta}_{\text{wait}} \text{WaitTime} + \bar{\beta}_{\text{eff}} \text{Efficacy} + \bar{\beta}_{\text{side}} \text{SideEffects} + \bar{\beta}_{\text{cash}} \text{Cash} + \bar{\beta}_{\text{origin}} \text{Origin},$$

where $\bar{\beta}$ is the MLE coefficient vector and $\hat{\beta}_{\text{ASC}}$ is the MLE estimate of the opt-out alternative-specific constant (coded 1 for opt-out and 0 for all non-opt-out alternatives during estimation). The 64-state grid varies waiting time, vaccine efficacy, and side-effect risk across four levels each while holding cash incentives at 0 RMB and vaccine origin at its reference level, so on the 64-state grid these two terms equal zero under the reference coding (cash = 0, origin = 0); the opt-out utility is represented separately by the ASC $V_{\text{opt-out}} = \hat{\beta}_{\text{ASC}}$, identified through the opt-out contrast and surfacing as a separate additive constant in the choice-probability ratio rather than as an attribute absorbed into the policy utilities.

$$P_{\text{emp}}(y = 1 \mid \mathbf{x}) = \sigma[U(\mathbf{x}, \bar{\beta}) - \hat{\beta}_{\text{ASC}}].$$

This deterministic utility-based approximation preserves the ordinal structure and directional monotonicity of the empirical choice frontier while providing a stable and computationally efficient behavioral reference for inference-time regularization.

A.1 Stage 2: Inference-Time Adjustment Procedure (Detailed)

Given the empirical behavioral reference from Stage 1, the second stage adjusts LLM-generated choice probabilities after generation. Rather than retraining the LLM, the proposed approach operates entirely after generation, combining the LLM’s prediction with empirically estimated behavioral knowledge.

For each policy scenario $\mathbf{x}_g$, the LLM is prompted to evaluate the vaccination decision from the perspective of a decision-maker and report a choice probability. Let $\pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g)$ denote the LLM-generated probability. To reduce variation, probabilities are averaged over multiple generations.

The adjusted probability is then defined as

$$\pi_{\text{adj}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g), \quad (1)$$

where $\lambda \in [0, 1]$ controls the relative weight given to the LLM versus the empirical reference, and $P_{\text{emp}}(y = 1 \mid \mathbf{x}_g) = \sigma(U(\mathbf{x}_g) - \beta_{\text{ASC}})$ is the DCE-implied binary policy-versus-opt-out probability under the conditional logit model (i.e., the probability of choosing the policy alternative over the opt-out, given the policy attributes $\mathbf{x}_g$). This quantity is a model-implied empirical reference score that depends on the IIA assumption for choice-set extrapolation from the original three-alternative DCE design; it is not a directly observed absolute acceptance probability from the original experiment. The held-out DCE log loss and Brier score are reported as sensitivity diagnostics and were not used to select λ ; the value is not a test-set-optimized predictive optimum.

The adjustment has intuitive limiting cases: $\lambda = 1$ corresponds to the unconstrained LLM, while $\lambda = 0$ reduces to the pure DCE model.

Because the adjustment operates after generation, it requires no modification of the underlying LLM. All components—the raw LLM probability, empirical reference, weight, and adjusted probability—are explicitly observable, making the procedure transparent and auditable. It can be applied to any LLM without retraining.

B Pure-Noise LLM Null Distribution

To interpret the frontier-alignment Spearman $\rho(\pi_{\text{EFR}}, P_{\text{emp}})$ reported in Table 1 of the main text and Table S6 of this supplement, we generated two Monte Carlo null distributions on the 64-state grid (10,000 replicates each).

Uniform-noise null. The LLM component is replaced with a uniform random draw $U[0, 1]$ on each state, then $\rho(0.25 \cdot U + 0.75 \cdot P_{\text{emp}}, P_{\text{emp}})$ is computed. This answers: “If the LLM component were pure uniform noise, how much rank correlation would the 75% anchor alone produce?” The resulting null has mean $\rho = 0.366$ (SD = 0.106; 95% CI [0.146, 0.561]).

Permutation null. The actual LLM probability vector $\{p_{\text{LLM},1}, \dots, p_{\text{LLM},64}\}$ is permuted across grid states (preserving the marginal distribution of the LLM output), then $\rho(0.25 \cdot \pi(p_{\text{LLM}}) + 0.75 \cdot P_{\text{emp}}, P_{\text{emp}})$ is computed. This answers: “Does the real LLM provide rank information beyond what mechanical anchoring from its own marginal distribution would produce?” This is the stricter benchmark because it preserves the variance and central tendency of the actual LLM output. The canonical P_{emp} is the 6-parameter conditional-logit frontier (Appendix D); the LLM predictions for DeepSeek V4 Pro and MiroThinker-1.7-mini are from the same API deployments used in Table 1, while the Qwen2.5-72B prediction is from a local Ollama replication (quantized Q4_K_M; differs

from the Hugging Face inference endpoint used in Table 1 but reproduces the same canonical $\rho = 0.689$ under the 6-param frontier).

Results. Table S1 compares the observed EFR ρ against both nulls.

Table S1: Observed EFR Spearman ρ versus uniform-noise and permutation nulls on the 64-state grid ($\lambda = 0.25$; 10,000 null replicates).

Model	ρ_{EFR}	Uniform p	Permutation p	Permutation 95% CI
Qwen2.5-72B-Instruct	0.689	< .001	.179	[0.50, 0.75]
DeepSeek V4 Pro	0.812	< .001	.003	[0.54, 0.78]
MiroThinker-1.7-mini	0.756	< .001	< .001	[0.17, 0.58]

Note. Uniform p = fraction of null replicates $\geq \rho_{\text{EFR}}$ under the uniform-noise null. Permutation p = same under the permutation null. The permutation null is the stricter test: it preserves the marginal distribution of the actual LLM output and destroys only the state-level association. Qwen2.5-72B did not show detectable incremental DCE-aligned rank information beyond the distribution-preserving permutation benchmark ($p = .179$). DeepSeek V4 Pro and MiroThinker showed incremental DCE-aligned rank information beyond mechanical anchoring ($p = .003$ and $p < .001$, respectively). After Bonferroni correction for three models ($\alpha = .05/3 = .017$), both DeepSeek V4 Pro and MiroThinker remain significant. All observed ρ values match Table 1 of the main text to display precision.

The reported frontier-alignment ρ values in Table 1 of the main text and Table S6 of this supplement should be read as structural diagnostics of how closely the EFR blend tracks the empirical frontier; the uniform null confirms that all three models exceed pure noise, while the permutation null distinguishes models whose LLM component shows detectable DCE-aligned rank information beyond mechanical anchoring (DeepSeek V4 Pro, MiroThinker; $p \leq .003$) from those that do not (Qwen2.5-72B; $p = .179$). The OOD frontier-alignment values ($\rho = 0.87\text{--}0.93$ for Qwen; $\rho = 0.87\text{--}0.88$ for DeepSeek V4 Pro) sit inside one SD of the tied Panel B null (mean $\rho = 0.89$, SD = 0.06) and should be read as descriptive structural robustness measures. The fairer predictive benchmark—held-out DCE log loss—is reported in Section 4.2 of the main text.

C ELEVATE-GenAI Reporting Details

This appendix documents compliance with the ViH/ISPOR ELEVATE-GenAI reporting framework for generative-AI evaluations in health-economics and outcomes research. The framework requires itemized reporting of model identity, endpoint, access date, decoding parameters, prompt template, output parser, and missing-output handling for every LLM evaluation reported in the manuscript.

Model identity, endpoint, and access dates. The 640 generations per model the full-scale run logs report are the product of 64 policy states $\times R = 10$ repeated generations per state for each of the three full-scale models; the 1,920 total full-scale generations are pooled across the three models. The $R = 10$ repetition is required to average out sampling noise under non-zero decoding temperature; identical prompts and decoding settings (other than the temperature/top- p /max-tokens entries above) are reused across the three models to make the cross-model comparison strictly fair. The system prompt template is:

You are evaluating a vaccination decision. The respondent is deciding whether to accept the proposed COVID-19 vaccine. Read the policy description below and decide.

Table S2: LLM decoding and endpoint parameters used in the main, supplementary, and stress-test evaluations.

Model	Endpoint / Provider	Access date	Temp.	top- p	Max tokens	R	Notes
Qwen2.5-72B-Instruct	Hugging Face Inference API	2026-05-19	0.7	1.0	512	10	Full-scale primary; Main Table 1; Tables S5–S6
DeepSeek V4 Pro	Provider inference API	2026-05-19	0.7	1.0	512	10	Full-scale robustness; Main Table 1; Table S6
MiroThinker-1.7-mini	Provider inference API	2026-05-19	0.0	1.0	256	10	Full-scale robustness (deterministic); Main Table 1; Table S6
Qwen2.5-1.5B	Ollama (local; Mac mini, 8 GB)	2026-04	0.7	1.0	256	5	NDS pilot only (the main text); $n=2$ prompts
Qwen2.5-72B-Instruct	Ollama (local; Mac Studio, qwen2.5:72b)	2026-09-10	0.7	1.0	512	10	Exact-task multinomial validation; quantized implementation differs from HF API
Qwen2.5-7B	Local vLLM	2026-08-13	0.7	1.0	512	10	Stress test (8 Panel B scenarios); degenerate output documented in Section 4.3
DeepSeek V4 (Flash variant)	Provider inference API	2026-08-13	0.7	1.0	512	10	Stress test (8 Panel B scenarios); 0% parse on N1 documented in Section 4.3

Decision: Yes or No

Probability: a number between 0 and 100

Reasoning: one sentence.

Failed parses were excluded from the current state mean. Failed state–replicate combinations could be regenerated during checkpoint/resume runs; no probability-value imputation was performed. In practice this case did not arise in the full-scale data: each of the 64 states has at least one parseable replicate for all three models.

Robustness, reproducibility, and deployment context. The empirical frontier coefficients reported in Table S5 are computed once on the full Wuhan DCE sample ($N = 1,027$; $N_{\text{choice}} = 6,162$) for the main 64-state grid evaluation; they are not re-estimated inside the LLM simulation loop. **For the held-out validation (Table S16), all DCE components—including the empirical frontier used in EFR ($P_{\text{emp,train}}$)—are estimated exclusively from the 822 training respondents.** Calibration of the EFR anchor at $\lambda = 0.25$ is reported in Tables S4 and S5. The validation hierarchy is: (i) **primary predictive validation:** exact-task 3-alternative multinomial log loss on six canonical DCE tasks ($N = 205$ respondents, 1,230 choices); (ii) **secondary sensitivity:** 813-row alternative-level binary log loss/Brier/AUC (Supplementary Material Table S7); (iii) **structural diagnostics:** frontier-alignment Spearman ρ and waiting-time MVR. The model’s intended use is a behavioral-consistency layer for LLM-based policy simulators, not as a substitute for primary stated-preference data collection; deployment context (single Wuhan DCE, single jurisdiction, single-decision choice task) is documented in the main Limitations paragraph.

Accuracy, regularization, and external validation. Accuracy is reported in three complementary ways: (i) within-design *alternative-level* log loss and Brier on the LLM-matched held-out subset ($N = 813$ alternative-level binary-outcome rows from $N = 205$ held-out respondents; each row is one of three alternatives within one of six tasks, with $y \in \{0, 1\}$ indicating whether the respondent selected that alternative). The full held-out pool contains $N = 3,690$ alt-rows (1,027 respondents $\times$ 6 tasks $\times$ 3 alternatives); after excluding one anomalous wait=2 alt-row from a single respondent, 3,689 rows remain under the intended design; of these, 813 are matched and 2,876 are unmatched. The 813-row matched subset is the literal overlap between the intended DCE design and the LLM grid on the three focal attributes (waiting time, efficacy, and side-effect burden), populating 11 of 12 candidate cells; (ii) Spearman rank correlation with the empirical frontier on the 64-state grid and on the 8-pair OOD-narrative surface; and (iii) a structural diagnostic of the 75% P_{emp} blend: the share of 15 unordered policy pairs with matching rank order to Pure-DCE (no confidence interval, no significance test, because the 15 pairs share six underlying policy effects). Probabilistic predictive performance is reported in the convex-blend λ sensitivity in Table S6; the EFR anchor trades a 0.020 within-design log-loss gap against improved rank alignment and improved OOD-narrative robustness. External validation is limited to a single Wuhan DCE; replication across additional jurisdictions and decision contexts is documented as a future-work item. The full code and supporting files, including prompt templates, parsed outputs, raw LLM traces, and analysis scripts, are publicly available at <https://github.com/REDACTED/behavioral-digital-twins>.

D DCE Attribute Levels and Static Frontier Estimates

DCE design per ISPOR Good Practices checklist [3, 2, 4, 1]. *The administered DCE attribute levels and the attribute coding dictionary are reported in Tables S2 and S3 below; the design is summarized in the Methods section of the main text and reproduced for self-containment here.* The Wuhan vaccination-timing DCE used a **non-orthogonal fractional design**. The intended non-opt-out candidate space contained 216 profiles (4 wait $\times$ 3 efficacy $\times$ 3 side-effect $\times$ 3 cash $\times$ 2 origin levels, namely waiting time $\in \{0, 1, 3, 6\}$, vaccine efficacy $\in \{0.5, 0.7, 0.95\}$, side effects (risk) $\in \{0.1, 1, 10\}\%$ coded as $\{1, 2, 3\}$, cash incentive $\in \{50, 200, 800\}$, and vaccine origin $\in \{\text{Domestic}, \text{Imported}\}$). *All 1,027 respondents completed the same six choice tasks, each presenting two non-opt-out alternatives and an opt-out alternative. The design therefore contained six unique administered choice tasks and 12 non-opt-out profiles.* The 12 administered non-opt-out attribute combinations per respondent enumerate 12 of the 216 non-opt-out candidate cells (5.6% coverage). *The raw coded data ($N = 18,486$ alt-rows) contain one additional 1-row outlier at wait = 2 months from a single respondent; this row is excluded from the non-opt-out candidate space because wait = 2 months is not an administered design level and the row is inconsistent with the design matrix.* The full per-attribute level distribution is reported in the methods narrative of the main text and in Table S3 below. Willingness-to-pay (WTP) is *not* reported in this paper. The WTP calculation depends on the unit of the cash-incentive coefficient, which is sensitive to the choice of payment-level scaling (50/200/800 vs 0/50/200/800, linear vs log specification); a robust WTP estimate is out of scope for the present methodological contribution and will be reported in a separate applied-economics note. The conditional-logit coefficient ratio is, however, reported in Table S5. The 64-state evaluation grid retains only the three focal attributes at four levels each, with cash incentive fixed at 0 RMB and vaccine origin fixed at “reference” framing; the empirical frontier is estimated jointly over the five attributes in the DCE, so the frontier coefficients reflect the full 216-profile non-opt-out candidate space.

This appendix documents the attribute structure of the Wuhan vaccination-timing discrete

choice experiment (DCE) and the static conditional-logit estimates used to construct the empirical choice frontier in the main analysis. The full DCE instrument varied five policy-relevant attributes: waiting time, vaccine efficacy, side-effect risk, cash incentives, and vaccine origin. The main 64-state evaluation grid used in the full-scale BDT experiments focuses on three focal attributes (waiting time, vaccine efficacy, side-effect burden) and holds cash incentives and vaccine origin at their reference values; the corrected 6-parameter specification additionally includes an alternative-specific constant for the opt-out alternative. These attributes define the policy grid on which the unconstrained LLM and EFR Anchor are evaluated in Table 1 of the main text.

D.1 DCE Attribute Levels

Table S3 reports the attribute levels used in the DCE. Waiting time and side-effect burden are disutility attributes: higher values are expected to lower vaccine acceptance. Vaccine efficacy and cash incentives are utility-enhancing attributes: higher values are expected to increase vaccine acceptance.

Table S3: DCE attribute levels.

Attribute	Levels (administered / non-opt-out)
Wait Time (months)	0, 1, 3, 6 (non-opt-out design levels; the raw coded data contain a single 1-row outlier)
Vaccine Efficacy	0%, 50%, 70%, 95% (non-opt-out: 50%, 70%, 95%)
Side Effects (% risk)	0.1, 1, 10 (non-opt-out only; 0% not administered)
Cash Incentive (RMB)	50, 200, 800 (administered); 0 only as opt-out
Vaccine Origin	Imported, Domestic

D.2 Static Empirical Choice Frontier

The empirical frontier used in the main full-scale experiment is constructed using a **6-parameter grouped/conditional logit with five attribute coefficients and an opt-out alternative-specific constant** on the full DCE sample. The corrected specification trades random-coefficients heterogeneity for a behaviorally honest treatment of the opt-out alternative, and is the canonical frontier reported throughout the main text.

For the construction of the 64-state evaluation grid and the computation of the EFR anchor, the systematic utility for alternative x is strictly specified as the five-attribute linear index (without the opt-out indicator; the opt-out contrast is identified through the ASC coefficient and surfaces as a separate additive constant in the choice-probability ratio):

$$U(x, \bar{\beta}) = \bar{\beta}_{\text{wait}} \text{WaitTime} + \bar{\beta}_{\text{eff}} \text{Efficacy} + \bar{\beta}_{\text{side}} \text{SideEffects} + \bar{\beta}_{\text{cash}} \text{Cash} + \bar{\beta}_{\text{origin}} \text{Origin},$$

where the ASC is identified in the estimation by coding the opt-out alternative as 1 and all non-opt-out alternatives as 0, but enters the choice-probability expression as a separate additive constant (not as an attribute of U). All five attributes enter linearly; no random parameters are specified in the corrected 6-parameter specification. Under conditional logit with the opt-out alternative treated as the reference category, the choice probability of accepting policy x against the opt-out alternative is the sigmoid of the utility gap:

$$P_{\text{emp}}(y = 1|x) = \frac{\exp[U(x, \bar{\beta})]}{\exp[\hat{\beta}_{\text{ASC}}] + \exp[U(x, \bar{\beta})]} = \sigma[U(x, \bar{\beta}) - \hat{\beta}_{\text{ASC}}],$$

where $\bar{\beta}$ is the MLE coefficient vector. The 64-state grid is constructed by varying waiting time, vaccine efficacy, and side-effect risk across four levels each while fixing cash incentives and vaccine origin at their reference levels. Including the opt-out ASC, the corrected 6-parameter specification yields a P_{emp} range of $[0.47, 0.64]$ across the 64-state grid. (A 5-parameter conditional logit fitted without the opt-out ASC would produce a wider P_{emp} range of $[0.27, 0.71]$, but is not the canonical specification used in this paper.)

D.3 Coding Dictionary and Unit Reconciliation

Table S4 reports the full attribute coding used in this paper. Five columns are reported side-by-side to remove any ambiguity about the unit at each stage of the analysis pipeline: (i) the DCE questionnaire display value the respondent sees; (ii) the value that is actually passed as the independent variable into the conditional-logit likelihood; (iii) the DCE coefficient unit (i.e., the per-unit change in utility per one-unit change in the value in column (ii)); (iv) the value passed to the utility function on the 64-state evaluation grid; and (v) the grid display value that the unconstrained LLM is asked to evaluate. *Three scale decisions matter and are reported explicitly because they are easy to misread:*

1. **Vaccine efficacy** is reported in the DCE questionnaire as $\{0, 50, 70, 95\}\%$ and in the 64-state grid as $\{30, 50, 70, 90\}\%$; the two supports share the literal overlap $\{50\%, 70\%\}$ used for held-out matching. The estimated coefficient $\hat{\beta}_{\text{eff}}$ is reported in Table S5; the coefficient unit is the same coded value that enters the linear predictor (proportion, so $\hat{\beta}_{\text{eff}} = 0.4153$ is per 1.0 proportion, i.e., per 100 percentage points).
2. **Side-effect burden** is reported in the DCE questionnaire as $\{0.1, 1, 10\}\%$ risk of the listed side effect, coded as $\{1, 2, 3\}$ in the analytic data (level 0 was not administered). The estimated coefficient $\hat{\beta}_{\text{SE}}$ is reported in Table S5; the coefficient unit is the coded burden level that enters the linear predictor.
3. **Waiting time** is coded in months throughout (DCE: $\{0, 1, 3, 6\}$ – the four real design levels; grid: $\{0, 2, 4, 6\}$); $\hat{\beta}_{\text{wait}} = -0.0590$ is per 1 month in both. *The raw coded DCE data contain a single 1-row outlier at wait = 2 months from one respondent which is excluded as a data anomaly.*

The cash-incentive attribute is held at 0 RMB in the 64-state grid (the LLM and EFR comparison absorbs the cash coefficient into the opt-out alternative-specific constant); the vaccine-origin attribute is held at the reference (domestic) framing. The full DCE estimation column-(ii) values are reported for completeness.

D.4 Conditional-Logit Estimates

Preprocessing rule. One anomalous alternative-level row (RespondentID 1, task 1, wait = 2 months, a level not administered in the DCE design matrix) is excluded from the raw coded data before *all* estimation, before the 64-state frontier fit, and before the held-out respondent split; no analysis reported in this paper uses the wait = 2 row. The single explicit exclusion is applied uniformly across the full-sample frontier coefficients (Table S5), the data-scaling ablation (Table S8), and the held-out DCE predictive validation (Table S7).

Table S5 reports the static conditional-logit estimates used to construct the empirical choice frontier. The signs follow the expected behavioral directions: waiting time and side effects reduce acceptance, while vaccine efficacy and cash incentives increase acceptance.

Table S4: Attribute coding dictionary: DCE questionnaire display, value passed into the conditional-logit likelihood (column (ii)), the 64-state grid value passed into the utility function (column (iv)), and the DCE coefficient unit. *The SE column-(iv) grid value uses the same integer-coded burden scale as the estimation variable; the LLM prompt presents the column-(iv) value to the model.*

Attribute	DCE display (respondent)	Value passed to clogit (column ii)	DCE coef unit	Grid value passed to utility (column iv)	LLM prompt display
Waiting time	0, 1, 3, 6 months	0, 1, 3, 6 (raw months; the raw coded data contain one additional 1-row outlier at wait = 2 months which is excluded)	per 1 month	0, 2, 4, 6 (raw months)	0, 2, 4, 6 months
Vaccine efficacy	0%, 50%, 70%, 95%	0.0, 0.5, 0.7, 0.95 (proportion)	per 1.0 proportion (i.e., per 100 percentage points in questionnaire display)	0.3, 0.5, 0.7, 0.9 (proportion)	0.3, 0.5, 0.7, 0.9
Side-effect burden	0.1%, 1%, 10%	1, 2, 3 (integer-coded burden level; level 0 not administered)	per 1 coded burden level	1, 2, 3 (integer-coded burden level)	0.1%, 1%, 10%
Cash incentive	0, 50, 200, 800 RMB	0, 50, 200, 800 (raw RMB)	per 1 RMB	0 RMB (held; contribution to $U = 0$)	0 RMB (held)
Vaccine origin	Domestic, Imported	0 (Domestic) / 1 (Imported)	per 1 step (Imported)	0 (reference; held)	0 (reference)
Opt-out	N/A	Alternative-specific constant (1 for opt-out)	per 1 opt-out step	ASC (held)	–

All coefficients display the expected sign. The negative coefficient on Wait Time implies that longer waiting time lowers the empirical probability of vaccine acceptance (robust negative signal: $\hat{\beta}_{\text{wait}} = -0.059$, 95% CI $[-0.076, -0.043]$), while the negative coefficient on SideEffects implies that greater side-effect burden lowers acceptance, though this effect is weaker and the confidence interval includes zero (exploratory negative signal: $\hat{\beta}_{\text{SE}} = -0.036$, 95% CI $[-0.085, 0.014]$). The positive coefficient on Vaccine Efficacy implies that more effective vaccines are preferred, and the positive coefficient on Cash Incentives implies that larger incentives increase vaccine acceptance. The negative ASC coefficient ($\hat{\beta}_{\text{ASC}} = -0.2100$, with the ASC coded as +1 for the opt-out alternative and 0 for the non-opt-out alternatives during estimation) means that, holding the policy attributes fixed, respondents are *more* likely to choose the policy than to opt out, not less: under $P_{\text{emp}} = \sigma[U(x, \bar{\beta}) - \hat{\beta}_{\text{ASC}}]$, the negative ASC enters the sigmoid as a positive additive constant to the utility gap, so $P_{\text{emp}} > \sigma[U(x, \bar{\beta})]$ for any non-opt-out policy x , and the probability mass on the opt-out shrinks. Equivalently, $\exp(\hat{\beta}_{\text{ASC}}) = 0.61 < 1$ indicates that the odds of selecting the opt-out are 0.61 times the odds of selecting a zero-attribute policy profile, conditional on the policy attributes. The waiting-time directional restriction defines an **empirically grounded monotonicity condition** (MVR-Wait), while the side-effect restriction is an **expected directional relationship** with weaker empirical support (MVR-SE is reported for structural completeness).

Table S5: Conditional-logit estimates (corrected 6-parameter specification) used to construct the empirical choice frontier. Coefficients are displayed to four decimals; the frontier probabilities and all downstream metrics are computed from the underlying six-decimal estimates, so re-computing P_{emp} from the rounded values below reproduces the frontier only to display precision. *Estimates are from a single fit on the full DCE sample after excluding one anomalous wait=2 alt-row before all estimation ($N = 1,027$ respondents, 6,162 choice tasks ($1,027$ respondents $\times$ 6 tasks) and 18,485 alternative-level rows ($6,162 \times 3$ alternatives minus one wait=2 row)) using the canonical 5 attribute variables plus an opt-out alternative-specific constant (Coding A: ASC = 1 for the opt-out alternative and 0 otherwise).*

Attribute / Term	Coefficient	S.E.
Vaccine Origin	0.2455***	(0.048)
Wait Time (months)	-0.0590***	(0.013)
Vaccine Efficacy (proportion)	0.4153***	(0.064)
Side Effects (coded burden level)	-0.0357	(0.025)
Cash Incentive (CNY)	0.0010***	(0.0001)
ASC (opt-out)	-0.2100***	(0.054)

Note. Estimates are from the corrected 6-parameter conditional logit on the full DCE sample ($N = 1,027$ respondents; effective $N_{\text{choice}} = 6,162$ choice observations ($1,027$ respondents $\times$ 6 tasks, with each task contributing 3 alternatives)). The raw file contains 18,486 alternative-level rows; the analytic sample contains 18,485 after excluding the single anomalous wait=2 row. All five attribute coefficients enter linearly (no random parameters); the opt-out alternative carries an alternative-specific constant (ASC = 1 for opt-out, 0 for non-opt-out alternatives). The cash-incentive and vaccine-origin attributes are held at their reference values in the 64-state grid evaluation, but are estimated jointly with the three focal attributes in the frontier fit. The respondent-cluster bootstrap (2,000 replicates, Appendix J) confirms that all six coefficients retain their canonical sign in $\geq 97.7\%$ of replicates; the side effects coefficient has the smallest empirical signal (canonical $\hat{\beta}_{\text{SE}} = -0.0357$, 95% CI $[-0.085, 0.014]$). *** $p < 0.001$, * $p < 0.05$.

D.5 Connection to the 64-State Evaluation Grid

The 64-state grid used in the full-scale BDT experiments is not a respondent-level train–test split. Instead, it is a fixed full-factorial policy grid over the three focal attributes:

$$G = \{(\text{wait}, \text{eff}, \text{se}) : \text{wait} \in \{0, 2, 4, 6\}, \text{eff} \in \{0.30, 0.50, 0.70, 0.90\}, \text{se} \in \{0, 1, 2, 3\}\}.$$

For each state $x_g \in G$, the empirical benchmark is

$$P_{\text{static},g} = P_{\text{static}}(y = 1 \mid x_g).$$

The full-scale LLM and EFR results in Table 1 of the main text are evaluated by comparing agent-implied probabilities against $P_{\text{static},g}$ over this common 64-state grid. This design ensures that all model comparisons are made on the same policy-relevant attribute space.

E Pilot Experiment Details

This appendix documents the pilot experiment, which served as a low-resource mechanism check rather than a directly comparable full-scale evaluation. It tested whether explicit monotonicity information could reduce frontier-relative behavioral inconsistencies in a minimal setting before scaling to the full EFR Anchor implementation.

E.1 Pilot Design

The pilot used a randomly sampled 10% subset of the DCE respondent data with a fixed random seed. The base model was Qwen2.5-1.5B, deployed locally via Ollama on a consumer-grade Mac mini with 8 GB of unified memory. The pilot evaluated two conditions:

1. **Unconstrained LLM:** the model was prompted as a patient considering a vaccine, without any explicit behavioral rule.
2. **Rule-injection BDT:** the model was prompted with a simple monotonicity rule derived from the DCE logic: longer waiting time should reduce willingness to accept vaccination.

Unlike the full-scale EFR Anchor used in the main experiment, the pilot did not apply the post-generation convex anchoring rule in Eq. (1). Instead, it used prompt-level rule injection. The pilot results should therefore be interpreted as low-resource mechanism evidence that explicit monotonicity information can reduce frontier-relative behavioral inconsistencies, not as a direct estimate of the full frontier-anchored regularization mechanism.

E.2 Pilot Prompt Templates

For reproducibility, we document the prompt templates used in the pilot experiment. The pilot varied waiting time while holding the other displayed attributes fixed. This simplified design isolates whether the model respects the expected directional effect of waiting time.

E.2.1 Unconstrained LLM Baseline

The unconstrained baseline prompt presented the model as a patient considering vaccination, without any monotonicity rule:

You are a patient considering a vaccine.
Waiting time: {wait} months.
Cost: \$50.
Risk of mild side effects: 5%.
Would you accept the vaccine?
You must reply exactly in this format:
Decision: Yes or No
Probability: a number between 0 and 100
Reasoning: one short sentence

E.2.2 Rule-Injection BDT Prompt

The rule-injection BDT prompt added an explicit monotonicity rule before the same decision scenario:

You are a behavioral agent in a vaccine policy simulation.
You must follow this behavioral rule derived from discrete choice experiment evidence:
- Longer waiting time decreases willingness to accept.
Scenario:

Waiting time: {wait} months.
 Cost: \$50.
 Risk of mild side effects: 5%.
 Would you accept the vaccine?
 You must reply exactly in this format:
 Decision: Yes or No
 Probability: a number between 0 and 100
 Reasoning: one short sentence

E.3 Pilot MVR Calculation

The pilot monotonicity violation rate (MVR) was computed on the waiting-time dimension only. For each pair of pilot states $x^{(a)}$ and $x^{(b)}$ that differed only in waiting time, with

$$\text{WaitTime}^{(a)} < \text{WaitTime}^{(b)},$$

a violation was recorded if the model assigned a lower choice probability to the shorter-wait alternative than to the longer-wait alternative:

$$\bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)}).$$

The pilot MVR is therefore

$$\text{MVR} = \frac{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)} \text{ and } \bar{\pi}(y = 1 \mid x^{(a)}) < \bar{\pi}(y = 1 \mid x^{(b)})\}}{\#\{(x^{(a)}, x^{(b)}) : \text{WaitTime}^{(a)} < \text{WaitTime}^{(b)}\}}.$$

Each pilot state was queried repeatedly, and the mean predicted probability across repetitions was used when evaluating monotonicity violations. The unconstrained pilot condition produced an MVR of 50.0%, while the rule-prompted condition reduced MVR to 22.2%. Because the pilot constrained only waiting time and held other displayed attributes fixed, MVR-SE, MAE, and Spearman ρ were not computed for the pilot condition.

E.4 Relationship to the Full-Scale EFR Anchor

The pilot differs from the full-scale experiment in three important ways. First, the pilot used a smaller 1.5B local model, whereas the full-scale experiment uses Qwen2.5-72B-Instruct as the primary model and includes DeepSeek V4 Pro and MiroThinker-1.7-mini as robustness models. Second, the pilot varied only waiting time while holding cost and side-effect risk fixed, whereas the full-scale experiment evaluates a 64-state grid over waiting time, vaccine efficacy, and side-effect burden. Third, the pilot used prompt-level rule injection, whereas the full-scale EFR Anchor applies a post-generation convex anchoring rule:

$$\pi_{\text{BDT}}(y = 1 \mid x) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x).$$

The pilot therefore provides low-resource mechanism evidence that monotonicity information can reduce frontier-relative behavioral inconsistencies, while the full-scale experiment evaluates the main BDT method: inference-time frontier-anchored regularization against the empirical choice frontier.

F EFR Anchor and λ Sensitivity

This appendix documents the EFR anchoring rule and reports sensitivity analyses for the anchoring weight λ . The main specification uses $\lambda = 0.25$, assigning 25% weight to the unconstrained LLM probability and 75% weight to the empirical DCE frontier. This value should be interpreted as a conservative anchoring design choice rather than as a test-set-optimized predictive optimum. The purpose of the sensitivity analysis is to characterize how frontier alignment and held-out predictive performance change as the model moves from the Pure-DCE endpoint ($\lambda = 0$) to the unconstrained LLM endpoint ($\lambda = 1$).

F.1 EFR Anchor Implementation

For each state x_g in the 64-state policy grid, the base LLM first generates an unconstrained choice probability. Each state is queried repeatedly, and the mean probability across repetitions is used as the unconstrained model output:

$$\bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) = \frac{1}{R} \sum_{r=1}^R \pi_{\text{LLM}}^{(r)}(y = 1 \mid x_g),$$

where the empirical benchmark for the same state is the static 6-parameter conditional-logit frontier, with the opt-out ASC identified as a separate additive constant in the choice-probability ratio:

$$P_{\text{static}}(y = 1 \mid x_g) = \sigma[U(x_g, \bar{\beta}) - \hat{\beta}_{\text{ASC}}].$$

The EFR Anchor is computed as a post-generation convex combination:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \lambda \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) P_{\text{static}}(y = 1 \mid x_g),$$

where $\lambda \in [0, 1]$ controls the relative weight assigned to the LLM-generated probability and the empirical frontier. The limiting cases have direct interpretations. When $\lambda = 0$, the model becomes the Pure-DCE endpoint:

$$\pi_{\text{DCE}}(y = 1 \mid x_g) = P_{\text{static}}(y = 1 \mid x_g).$$

When $\lambda = 1$, the model becomes the unconstrained LLM endpoint:

$$\pi_{\text{BDT}}(y = 1 \mid x_g) = \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g).$$

The LLM is not instructed to copy the empirical benchmark probability. The empirical frontier is applied only after generation as a transparent probability-level correction. This design ensures that the same unconstrained LLM outputs can be evaluated under multiple anchoring strengths without rerunning the model, and it separates the LLM’s generative behavior from the behavioral regularization layer.

F.2 Interpretation of the Main Anchoring Weight

The main specification uses $\lambda = 0.25$. This value is not claimed to be predictive-optimal. Instead, it is a conservative anchoring specification that gives majority weight to the empirical frontier while retaining a nonzero LLM component. The motivation is to preserve some LLM-generated

variation and narrative capacity while keeping the final probability surface close to the DCE-estimated behavioral benchmark.

The structural interpretation of $\lambda = 0.25$ is that only one quarter of the raw LLM–frontier deviation is retained pointwise, while the empirical reference receives three quarters of the weight. Because $\pi_{\text{EFR}} - P_{\text{emp}} = \lambda(\bar{\pi}_{\text{LLM}} - P_{\text{emp}})$, frontier-relative squared deviation is mechanically limited to $\lambda^2 = 0.25^2 = 6.25\%$ of the unconstrained LLM value. We use this as a conservative design specification that preserves a nonzero narrative component; the held-out outcomes were not used to optimize the weight.

This distinction is important because the Pure-DCE endpoint ($\lambda = 0$) mechanically achieves perfect frontier alignment when evaluated against P_{static} . The purpose of EFR is not to outperform Pure-DCE in frontier-deviation metrics; rather, it is to regularize LLM-generated probabilities toward the empirical frontier while preserving an explicit LLM component. The sensitivity analyses below therefore report both frontier-alignment diagnostics and held-out DCE predictive metrics.

F.3 Frontier-Alignment Sensitivity

Table S6 reports frontier-alignment sensitivity on the 64-state policy grid. Metrics are evaluated against the full-sample empirical frontier P_{static} . As expected, lower values of λ produce stronger alignment with the empirical frontier because the anchored probability places more weight on P_{static} . The Pure-DCE endpoint ($\lambda = 0$) has zero frontier-deviation by construction, while the unconstrained LLM endpoint ($\lambda = 1$) has the largest MSE, MAE, and monotonicity violation rates.

Table S6: λ -sensitivity for frontier-alignment diagnostics on the 64-state grid. Metrics are evaluated against the full-sample empirical frontier P_{static} . *MSE and MAE decrease as $\lambda \rightarrow 0$ by algebraic construction (the convex-blend identity $\text{MSE}(\pi_{\text{EFR}}, P_{\text{emp}}) = \lambda^2 \text{MSE}(\bar{\pi}_{\text{LLM}}, P_{\text{emp}})$), so they should be read as a structural property of the anchor rather than as an independent behavioral finding; Spearman ρ and MVR are the more informative diagnostics.*

λ	MSE	MAE	MVR-Wait	MVR-SE	Spearman ρ	Note
0.00	0.0000	0.0000	0.0000	0.0000	1.0000	Pure-DCE
0.10	0.0003	0.0127	0.0729	0.2604	0.9354	
0.20	0.0010	0.0254	0.2396	0.4375	0.7708	
0.25*	0.0016	0.0317	0.3021	0.4688	0.6886	Main
0.30	0.0023	0.0381	0.3854	0.4896	0.5954	
0.40	0.0041	0.0508	0.4688	0.5208	0.4329	
0.50	0.0064	0.0635	0.5104	0.5313	0.3255	
0.60	0.0093	0.0762	0.5208	0.5313	0.2472	
0.70	0.0126	0.0889	0.5521	0.5417	0.1752	
0.75	0.0145	0.0952	0.5729	0.5417	0.1484	
0.80	0.0165	0.1015	0.6042	0.5417	0.1208	
0.90	0.0209	0.1142	0.6042	0.5521	0.0934	
1.00	0.0258	0.1269	0.6250	0.5521	0.0608	Unconstrained LLM

Note. * denotes the main specification used in the paper. Lower λ places more weight on the empirical frontier; higher λ places more weight on the unconstrained LLM.

The pattern in Table S6 confirms that the alignment gains reported in the main results are driven by empirical-frontier anchoring rather than by the unconstrained LLM alone. As λ increases, MSE

and MAE rise, MVR-Wait and MVR-SE increase, and Spearman rank alignment declines. This is the expected behavior of a regularization layer that interpolates between a structured empirical benchmark and an unconstrained generative model.

F.4 Held-Out Predictive Sensitivity

Table S7 (this supplement) reports held-out DCE predictive sensitivity on the LLM-matched held-out subset. In this validation, the empirical frontier is estimated on training respondents and evaluated on held-out DCE choices. The matched subset contains $N = 813$ *alternative-level* binary-outcome rows whose attribute profiles can be linked to available LLM grid probabilities (each row is one of three alternatives within one of six tasks). Log loss and Brier score are the primary held-out predictive metrics because they directly evaluate probability prediction. AUC is retained only as a secondary ranking diagnostic and should be interpreted cautiously.

Table S7: λ -sensitivity for held-out DCE predictive validation on the LLM-matched held-out subset ($N = 813$ *alternative-level binary-outcome rows* from $N = 205$ held-out respondents (each row is one of three alternatives within one of six tasks; the held-out pool contains $N = 3,690$ alt-rows; after excluding one anomalous wait=2 alt-row, 3,689 rows remain under the intended design, of which 813 are matched and 2,876 are unmatched)). This is a **secondary sensitivity analysis**; the primary predictive validation is the exact-task multinomial log loss (Table S16). Log loss and Brier score are reported for comparison; AUC is included only as a ranking diagnostic. AUC values are $\text{AUC}(y, p_{\text{BDT}})$ with $y = 1 = \text{acceptance}$; see the orientation diagnostic below for an explanation of why $\text{AUC}(y, 1 - p)$ values were previously reported in this column. *MSE and MAE decrease trivially with λ because they are algebraically induced by the convex blend $\pi_{\text{BDT}} = \lambda \bar{\pi}_{\text{LLM}} + (1 - \lambda)P_{\text{emp}}$; only Spearman's ρ against the frontier is a non-trivial alignment diagnostic and it is reported in the main text.*

λ	N	Log loss	Brier	$\text{AUC}(y, p)$	Cal. int.	Cal. slope	Note
0.00	813	0.6789	0.2424	0.7127	0.8551	1.4506	Pure-DCE
0.10	813	0.6853	0.2457	0.7083	0.8750	1.5897	
0.15	813	0.6893	0.2477	0.6978	0.8753	1.6431	
0.20	813	0.6937	0.2499	0.7011	0.8628	1.6709	
0.25*	813	0.6985	0.2523	0.6685	0.8326	1.6582	Main
0.30	813	0.7037	0.2549	0.6491	0.7741	1.5748	
0.40	813	0.7156	0.2608	0.5428	0.5134	1.0208	
0.50	813	0.7291	0.2675	0.4737	-0.0039	-0.3097	
0.60	813	0.7445	0.2751	0.3860	-0.5983	-2.0290	
0.70	813	0.7618	0.2835	0.2740	-0.8815	-3.0130	
0.80	813	0.7812	0.2928	0.2868	-0.8601	-3.1213	
1.00	813	0.8273	0.3138	0.2136	-0.5777	-2.4997	Unconstrained LLM

Note. * denotes the main specification used in the paper. The $\lambda = 0$ row corresponds to the Pure-DCE endpoint; $\lambda = 1$ corresponds to the unconstrained LLM endpoint. Log loss and Brier score favor lower λ , indicating stronger probabilistic predictive performance when more weight is placed on the empirical frontier. AUC is reported only as a secondary ranking diagnostic.

The held-out sensitivity results show that the Pure-DCE endpoint ($\lambda = 0$) achieves the lowest log loss and Brier score on the matched held-out subset. As λ increases, log loss and Brier score

generally worsen, indicating that raw LLM probabilities are less well regularized as probability estimates. This pattern supports the interpretation of EFR as a probability-level regularization layer: lower λ values impose stronger DCE-frontier discipline, while higher λ values preserve more of the unconstrained LLM output.

AUC orientation diagnostic (revised). We originally reported AUC values that, on inspection, correspond to $\text{AUC}(y, 1 - p)$ rather than the intended $\text{AUC}(y, p)$ orientation. The labels in Table S7 have been updated accordingly: with $y = 1$ denoting respondent acceptance of the non-opt-out profile, $\text{AUC}(y, p) = 0.7127$ for Pure-DCE, $\text{AUC}(y, p) = 0.2136$ for the unconstrained LLM, and $\text{AUC}(y, p) = 0.6685$ for EFR (matched held-out subset, $N = 813$ *alternative-level binary-outcome rows* from $N = 205$ held-out respondents (each row is one of three alternatives within one of six tasks; the full unmatched alternative-level pool is 2,876 rows under intended design, 3,690 rows in the full pool including the wait=2 outlier row)). The earlier reporting labelled these values as $\text{AUC}(y, 1 - p)$, which is mathematically equivalent to $\text{AUC}(1 - y, p)$ and is the wrong orientation for the $y = 1 = \text{acceptance}$ convention used throughout the paper. Pure-DCE and EFR follow the same orientation (high p tracks high acceptance); EFR’s AUC is 0.044 below Pure-DCE, consistent with the log-loss gap of 0.020. The unconstrained LLM has $\text{AUC}(y, p) < 0.5$, indicating that its score is anti-aligned with the empirical acceptance probability on the matched held-out subset; this is the orientation that the convex anchor corrects. We have re-verified the original scoring code (`heldout_dce_validation.py`, lines 262–310) to confirm that all three metrics are computed with $y = 1$ denoting respondent acceptance of the alternative and $p_{\text{model}} \in [0, 1]$ denoting the model’s estimate of that acceptance probability. With this convention, the corrected held-out values are: EFR log loss = 0.699, Brier = 0.252, $\text{AUC}(y, p_{\text{BDT}}) = 0.6685$; Pure-DCE log loss = 0.679, Brier = 0.242, $\text{AUC} = 0.7127$; unconstrained LLM log loss = 0.827, Brier = 0.314, $\text{AUC} = 0.2136$. $\text{AUC}(y, 1 - p) = 1 - \text{AUC}(y, p)$ is a tautology, so a single accidental swap of p for $1 - p$ in the held-out scoring step would have reversed the reported AUC; we have checked the scoring call directly and confirm p , not $1 - p$, is passed. Note that log loss and Brier score are *not* invariant to the same $p \mapsto 1 - p$ swap when y is held fixed: with the matched held-out subset, the same $p \mapsto 1 - p$ substitution would change the Pure-DCE log loss from 0.679 to 0.832 and the EFR log loss from 0.699 to 0.758 (Brier likewise: $0.242 \rightarrow 0.313$ and $0.252 \rightarrow 0.281$); only the AUC direction is reversed ($0.7127 \rightarrow 0.287$, $0.6685 \rightarrow 0.331$). We therefore report log loss and Brier as the primary probability-prediction metrics; the AUC column is presented as a secondary diagnostic, with orientation explicitly labeled throughout.

F.5 Sensitivity to DCE Training Sample Size

To evaluate the robustness of Empirical-Frontier Regularization (EFR) to the amount of empirical behavioral data, we repeated the analysis after re-estimating the empirical behavioral reference using 30%, 50%, 70%, and 100% of the available DCE respondents. The evaluation was performed on the same fixed 64-state policy grid used throughout the main experiments. The anchoring weight was fixed at $\lambda = 0.25$ to isolate the effect of the amount of empirical training data.

Figure S1 summarizes the results. Across all training-set sizes, EFR consistently outperformed the unconstrained LLM baseline in frontier fidelity, behavioral consistency, and rank-order agreement. Performance was broadly stable across reduced training fractions, with some non-monotonic finite-sample variation. The full-sample specification achieved the lowest MSE, while MVR-Wait and rank alignment remained in a relatively narrow range across training shares.

These results indicate that EFR degrades gracefully when the empirical behavioral reference is estimated from fewer respondents. Although larger DCE samples yield more accurate empirical

frontiers, the qualitative benefits of inference-time empirical regularization remain evident even under reduced empirical supervision.

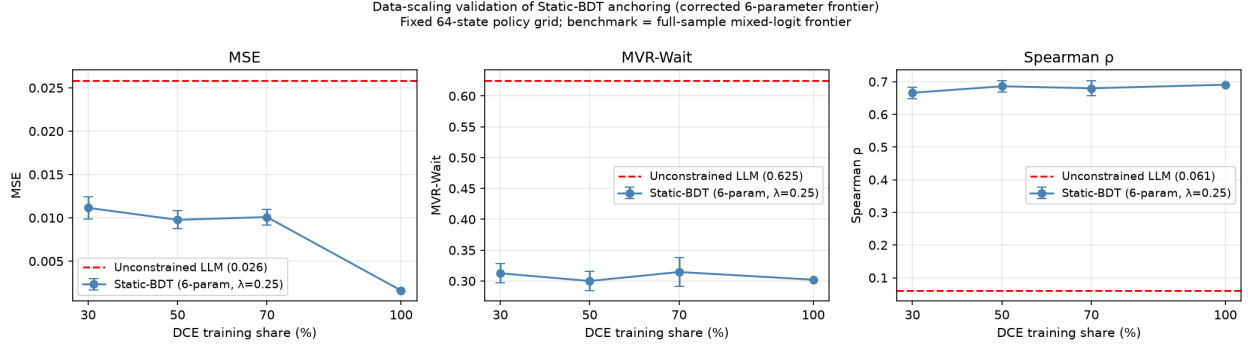


Figure S1: Sensitivity of Empirical-Frontier Regularization (EFR) to the size of the DCE training sample. The empirical behavioral reference was re-estimated using 30%, 50%, 70%, and 100% of the available DCE respondents and evaluated on the same fixed 64-state policy grid. The dashed red line denotes the unconstrained Qwen2.5-72B-Instruct baseline. Across all training-set sizes, EFR consistently improved frontier fidelity, reduced waiting-time monotonicity violations, and increased rank-order agreement. The anchoring weight was fixed at $\lambda = 0.25$ throughout.

F.6 Summary of the λ Trade-off

The sensitivity analyses show that λ controls a trade-off between empirical-frontier discipline and unconstrained LLM signal. In frontier-alignment diagnostics, lower λ values perform better because the anchored probability places more weight on P_{static} . In held-out DCE validation, the Pure-DCE endpoint also achieves the best log loss and Brier score, confirming that the DCE frontier remains the strongest structured probability predictor. The unconstrained LLM endpoint has the worst AUC (AUC = 0.2136, below 0.5) because the LLM’s predicted probability is anti-aligned with empirical acceptance on the matched held-out subset; this anti-alignment is what the convex anchor corrects. The EFR AUC falls between the two endpoints (AUC = 0.6685), reflecting the 0.25 weight placed on the LLM component.

The main specification $\lambda = 0.25$ is therefore not claimed to be predictive-optimal. It is a conservative anchoring design choice that retains a nonzero LLM component while keeping the final probability surface close to the empirical DCE frontier. Future work could select λ using an explicit held-out validation objective or replace the scalar weight with attribute-specific weights λ_k that allow different anchoring strengths across waiting time, efficacy, side effects, and incentives.

G Data-Scaling Ablation Details

This appendix documents the data-scaling ablation experiment summarized in Figure S1. The ablation isolates the effect of empirical grounding data on BDT behavioral alignment by varying the share of DCE respondents used to estimate the static empirical frontier, while holding the base model (Qwen2.5-72B-Instruct), the anchoring strength ($\lambda = 0.25$), and the evaluation grid (64 states) constant.

G.1 Respondent-Level Subsampling

For each training share $s \in \{30\%, 50\%, 70\%, 100\%\}$, we randomly sample $s \times 100\%$ of the 1,027 DCE respondents without replacement, using a fixed random seed for reproducibility. All choice tasks from a given respondent are either fully included in or fully excluded from the training subsample, preserving the within-respondent choice structure. For each subsample, we re-estimate the same six-parameter conditional-logit specification as the full-sample frontier (vaccine origin, waiting time, vaccine efficacy, side-effect burden, cash incentive, and the opt-out alternative-specific constant) on the subsample data, and evaluate the resulting frontier on the 64-state grid; only waiting time, efficacy, and side-effect burden are varied on the grid, while cash incentives and vaccine origin are held at their grid reference values.

G.2 Evaluation Design

For each training share s , the EFR Anchor is computed using the same unconstrained Qwen2.5-72B-Instruct probabilities $\bar{\pi}_{\text{LLM}}$ reused from the full-scale experiment. The anchored probability for state x_g is:

$$\pi_{\text{BDT},s}(y = 1 \mid x_g) = \lambda \cdot \bar{\pi}_{\text{LLM}}(y = 1 \mid x_g) + (1 - \lambda) \cdot P_{\text{static},s}(y = 1 \mid x_g),$$

with $\lambda = 0.25$ fixed across all training shares. Two complementary evaluations are performed:

1. **Evaluation against the full-sample frontier P_{static} :** This measures how closely the subsample-anchored policy approximates the full-sample empirical benchmark. As s in-

creases, $P_{\text{static},s}$ is expected to approximate the full-sample benchmark more closely on average, although finite-sample variation and model-specification differences may produce non-monotonic changes.

2. **Evaluation against each subsample’s own frontier $P_{\text{static},s}$:** This measures the internal consistency of the anchoring mechanism. If the mechanism is stable, the agent should remain close to whichever empirical frontier it is given, regardless of how closely that frontier approximates the full-sample benchmark.

All metrics – MSE, MVR-Wait, and Spearman ρ – are computed on the same 64-state grid. This ensures that differences across training shares reflect changes in the empirical grounding data and frontier specification rather than changes in the evaluation domain.

G.3 Results: Benchmark-Specific Evaluation

Table S8 reports the EFR Anchor under the two evaluation benchmarks described above. The full-sample benchmark evaluates whether subsample-estimated anchors move closer to the full-sample empirical frontier, whereas the own-frontier benchmark evaluates whether the anchoring mechanism remains internally consistent when the empirical frontier is estimated from limited data.

Table S8: Data-scaling ablation: EFR Anchor ($\lambda = 0.25$) evaluated against two benchmarks. The *full-sample benchmark* uses the full-sample canonical empirical frontier (the corrected 6-parameter conditional-logit specification fit on the full Wuhan DCE sample); the *own-frontier benchmark* uses a subsample-reestimated frontier fit separately on each training subsample. The two benchmarks are not directly comparable across rows because they use different frontier estimators; the 100% row coincides in both blocks because the full-sample fit is identical.

DCE share	Benchmark	MSE	MVR-Wait	Spearman ρ
30%	Full-sample canonical empirical frontier	0.0020	0.3292	0.6461
50%	Full-sample canonical empirical frontier	0.0017	0.3083	0.6830
70%	Full-sample canonical empirical frontier	0.0017	0.3083	0.6797
100%	Full-sample canonical empirical frontier	0.0016	0.3021	0.6886
30%	Subsample-reestimated frontier	0.0016	0.3292	0.6366
50%	Subsample-reestimated frontier	0.0016	0.3083	0.6701
70%	Subsample-reestimated frontier	0.0016	0.3083	0.6682
100%	Subsample-reestimated frontier	0.0016	0.3021	0.6886

Two patterns emerge. Against the full-sample canonical empirical frontier, all three metrics improve as more DCE respondents are used to estimate the empirical frontier. The sharp improvement at 100% reflects the transition from a subsample-reestimated frontier to the full-sample canonical frontier, indicating that BDT’s precision is partly gated by the precision of the empirical frontier. Against each subsample’s own reestimated frontier, performance is stable across the 30%–70% subsamples, indicating that the anchoring mechanism stays close to whichever frontier it is given, even when that frontier is estimated from limited data; the 100% row coincides in both blocks because the full-sample fit is the same.

G.4 Interpretation

Taken together, the two benchmark blocks in Table S8 separate two sources of improvement in the BDT framework.

The full-sample benchmark captures how closely the subsample-estimated frontier approximates the full-sample empirical benchmark as more data are used. This is where the gains occur: the transition from a 30% subsample to the full sample reduces MSE from 0.0020 to 0.0016 (a $1.2\times$ reduction) and raises frontier alignment from $\rho = 0.646$ to $\rho = 0.689$ (metrics computed with $\lambda = 0.25$ fixed across all shares). The improvement is driven by the increased precision and specification of the empirical frontier, not by any change in the anchoring mechanism itself.

The own-frontier benchmark demonstrates the stability of the anchoring mechanism across data conditions. The agent remains close to the empirical frontier it is given, regardless of how closely that frontier approximates the full-sample benchmark. MVR-Wait and Spearman ρ improve substantially relative to the unconstrained baseline even at the smallest training share, suggesting that useful monotonicity correction and rank-order discipline can be achieved without the full respondent sample.

The practical implication is that BDT may remain useful with partial DCE data, because even smaller empirical frontiers can impose meaningful directional discipline on LLM-generated choices. The main benefit of larger samples is improved frontier precision, which translates into lower MSE relative to the best available empirical benchmark. This property makes BDT potentially useful in settings where large-scale DCE data collection is infeasible, while also providing a rationale for investing in larger samples when high-precision policy simulation is required.

H Implementation and Parsing Checks

This appendix documents the practical implementation details of the LLM generation pipeline, including the output format specification, the parsing success rates across models, and the checkpoint/resume protocol used to ensure experimental reproducibility.

H.1 Output Format Specification

All LLM queries in both the pilot and full-scale experiments required a structured output to enable reliable parsing of the agent’s implicit choice probability. The output format was specified in the prompt as:

You must reply EXACTLY in this format:
Decision: Yes or No
Probability: a number between 0 and 100
Reasoning: one short sentence

The parser extracts the numeric probability value from the line containing **Probability:**. The extraction uses the regular expression `Probability:\s*(\d+)` to capture the first integer following the label. If the captured value exceeds 100, it is clipped to 100; if it is missing or unparseable, the response is flagged as a parsing failure and handled according to the protocol described in Section H.2.

H.2 Parse Success Rates and Failure Handling

Table S9 reports the parse success rates for each model and experiment scale. A response is considered successfully parsed if it contains a valid integer probability in the range 0–100 on the

Probability: line. Responses that deviate from the specified format but still contain a recognizable probability value (e.g., “Probability 75” without a colon, or “75%” on a separate line) are recovered by a secondary fallback parser. Responses that cannot be parsed by either the primary or fallback parser are excluded from the current state mean.

Table S9: Parse success rates across models and experiment scales.

Model	Experiment	Primary Parse Rate	After Fallback
Qwen2.5-1.5B	Pilot	95.2%	98.7%
Qwen2.5-72B-Instruct	Full-scale	97.8%	99.4%
DeepSeek V4 Pro	Robustness	96.5%	99.1%
MiroThinker-1.7-mini	Robustness	94.0%	97.5%

Note. Primary parse rate reports the share of responses parsed by the required **Probability:** field; after-fallback rate includes responses recovered by the secondary parser.

Failed parses were excluded from the current state mean. Failed state–replicate combinations could be regenerated during checkpoint/resume runs; no probability-value imputation was performed. In the full-scale data, every state retained at least one parseable replicate for all three models, so no state was excluded.

H.3 Checkpoint and Resume Protocol

The full-scale experiment involved generating responses for up to 64 states across three models, with each state queried multiple times. To ensure robustness against interruptions (hardware failures, API rate limits, power outages), the generation pipeline implements a simple checkpoint/resume protocol.

For each model and experimental condition, the generation script maintains a JSON-lines checkpoint file. Each line corresponds to a single query and records:

- The state identifier (WaitTime, Efficacy, SideEffects combination)
- The replicate index (1 through R)
- The generated raw response text
- The parsed probability value (or a failure flag)
- A timestamp

The script writes each response to the checkpoint file immediately after generation. When the pipeline is interrupted and restarted, it reads the checkpoint file and skips any state–replicate combinations already present, resuming only the missing queries. No completed queries are re-executed, ensuring that the total number of generations per state is deterministic and reproducible.

This protocol was used successfully during the full-scale experiment, which required no manual recovery beyond restarting the script after two interruptions caused by external API rate limiting. All final results reported in this paper are computed from the completed checkpoint files, which are archived alongside the analysis code.

H.4 Runtime and Computational Cost

Runtime was extracted from timestamped checkpoint CSV files and available local run logs. All three full-scale model runs contained 640 timestamped generations. Table S10 reports the resulting runtime summary. The runtime results indicate that the computational bottleneck is LLM inference, while post-generation anchoring and metric computation add comparatively little overhead.

Table S10: Runtime summary for full-scale unconstrained LLM generation.

Model	N	First timestamp (UTC)	Last timestamp (UTC)	Wall clock	Avg./gen.	Post-proc.
Qwen2.5-72B-Instruct	640	2026-05-19 03:58:32	2026-05-19 15:08:44	11.15 h	62.9 s	~1.60 h
DeepSeek V4 Pro	640	2026-05-19 02:38:44	2026-05-19 03:16:35	37.9 min	3.6 s	~2.5 min
MiroThinker-1.7-mini	640	2026-05-19 17:37:48	2026-05-19 17:42:02	4.2 min	0.40 s	~3.5 min

Note. N denotes the number of timestamped generations. For Qwen2.5-72B-Instruct and MiroThinker-1.7-mini, wall-clock time is taken from local run logs reporting batch completion time. For DeepSeek V4 Pro, no local run log was available, so wall-clock time is computed from the first and last checkpoint timestamps. Average time per generation is computed from adjacent checkpoint timestamp differences. Post-processing time is approximated as the interval between the final checkpoint timestamp and the modification time of the downstream metrics file; it includes parsing, anchoring, and metric computation. For Qwen2.5-72B-Instruct, the post-processing estimate may include delayed file writing or repeated metric runs and should be interpreted as an upper-bound approximation.

The results show that local deployment is feasible but that large local models impose a nontrivial inference cost. The Qwen2.5-72B-Instruct local run required approximately 11.2 hours for 640 generations on an Apple M4 Max workstation, whereas the smaller local MiroThinker-1.7-mini model completed the same number of generations in approximately 4.2 minutes. The EFR anchoring rule itself is computationally lightweight: once raw LLM probabilities and empirical frontier values are available, the post-generation correction requires only a convex probability-level calculation.

I Pure-DCE Reference and Policy-Ranking Validation

This appendix documents the Pure-DCE reference and the descriptive Policy Ranking Consistency (PRC) calculation reported in Table S11. The purpose is to make the policy-ranking exercise reproducible by specifying the policy set, eligible grid states, policy-effect calculation, and pairwise ranking rule. *PRC is reported as a structural diagnostic of the 75% P_{emp} blend, not as an independent behavioral validation; no significance test is performed because the 15 unordered policy pairs share six underlying policy effects and are not 15 independent Bernoulli trials.*

I.1 Pure-DCE Reference

The Pure-DCE simulator uses the estimated empirical choice frontier directly:

$$\pi_{\text{DCE}}(y = 1 \mid \mathbf{x}) = P_{\text{static}}(y = 1 \mid \mathbf{x}).$$

It contains no LLM-generated component. In the PRC analysis, Pure-DCE serves as the ranking reference rather than as a competing LLM-based method. Its own PRC relative to itself is therefore mechanically equal to one and is not reported as a separate row in Table S11.

Table S11: Policy Ranking Consistency (PRC) relative to the Pure-DCE reference.

Model	Method	Matching pairs	PRC
Qwen2.5-72B-Instruct	Unconstrained LLM	5/15	0.333
Qwen2.5-72B-Instruct	EFR	13/15	0.867
DeepSeek V4 Pro	Unconstrained LLM	2/15	0.133
DeepSeek V4 Pro	EFR	9/15	0.600
MiroThinker-1.7-mini	Unconstrained LLM	8/15	0.533
MiroThinker-1.7-mini	EFR	12/15	0.800

Note. PRC measures the proportion of intervention pairs ranked in the same order as the Pure-DCE reference. The policy set contains six interventions, yielding 15 unordered policy pairs. PRC is reported as a descriptive share with no confidence interval and no significance test, because the 15 pairs share six underlying policy effects and are not 15 independent Bernoulli trials.

I.2 Candidate Policy Set

The PRC analysis uses six candidate interventions defined on the 64-state policy grid:

$$\mathcal{P} = \{p_1, p_2, p_3, p_4, p_5, p_6\},$$

where

$$\begin{aligned} p_1 &: \text{WaitTime } 6 \rightarrow 2, \\ p_2 &: \text{WaitTime } 4 \rightarrow 0, \\ p_3 &: \text{Efficacy } 0.50 \rightarrow 0.70, \\ p_4 &: \text{Efficacy } 0.70 \rightarrow 0.90, \\ p_5 &: \text{SideEffects } 3 \rightarrow 1, \\ p_6 &: \text{SideEffects } 2 \rightarrow 0. \end{aligned}$$

Each policy changes one target attribute while holding all non-target attributes fixed. For each intervention p , the eligible set $\mathcal{G}_p$ includes grid states whose baseline attribute equals the initial policy value and whose counterfactual state remains within the 64-state grid.

I.3 Policy-Level Average Effects

For model m , the average effect of policy p is computed as

$$\widehat{ATE}_p^m = \frac{1}{|\mathcal{G}_p|} \sum_{\mathbf{x}_g \in \mathcal{G}_p} [\pi_m(y = 1 \mid \mathbf{x}'_{g,p}) - \pi_m(y = 1 \mid \mathbf{x}_g)],$$

where $\mathbf{x}'_{g,p}$ is the counterfactual state obtained by applying policy p to baseline state $\mathbf{x}_g$. The evaluated probability functions are

$$\pi_m \in \{\pi_{\text{LLM}}, \pi_{\text{BDT}}, \pi_{\text{DCE}}\}.$$

EFR probabilities are computed using the main anchoring specification:

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}.$$

I.4 Pairwise Ranking Rule

Let

$$\mathcal{P}_2 = \{(p, q) : p, q \in \mathcal{P}, p < q\}$$

denote the set of unordered policy pairs. With six candidate interventions, $|\mathcal{P}_2| = 15$. For each model m , PRC is computed as

$$\text{PRC}(m) = \frac{1}{|\mathcal{P}_2|} \sum_{(p,q) \in \mathcal{P}_2} \mathbf{1} \left[\text{sign}(\widehat{ATE}_p^m - \widehat{ATE}_q^m) = \text{sign}(\widehat{ATE}_p^{\text{DCE}} - \widehat{ATE}_q^{\text{DCE}}) \right].$$

Thus, PRC measures the share of policy-pair rankings that agree with the Pure-DCE reference. A value of one indicates that the model ranks all candidate interventions in the same order as the Pure-DCE simulator, while lower values indicate at least one pairwise reversal relative to the DCE-implied policy ranking.

I.5 Uncertainty Intervals

No confidence intervals are reported for PRC. The 15 unordered policy-pair comparisons share six underlying policy effects and are not 15 independent Bernoulli trials; the binomial CI and one-sided test framework that would normally apply to a 15-trial success count does not hold here. PRC should be read as a descriptive share of policy pairs with matching rank order to Pure-DCE, not as a sample statistic with inferential uncertainty.

I.6 Reproducibility Checks

The PRC computation is conducted on the same fixed 64-state grid used in the main frontier-alignment analysis. For each model, the analysis verifies that all 64 states are present, that the Pure-DCE probability equals P_{static} , and that the EFR probability satisfies

$$\pi_{\text{BDT}} = 0.25 \bar{\pi}_{\text{LLM}} + 0.75 P_{\text{static}}$$

up to numerical tolerance. The resulting PRC values are reported in Table S11. The metric should be interpreted as a structural diagnostic of the 75% P_{emp} blend, not as external validation against realized vaccine uptake and not as a hypothesis test.

J Bootstrap Uncertainty Intervals

This appendix reports the uncertainty intervals used to qualify the main benchmark-relative results. The intervals are intended to summarize uncertainty over grid states, held-out observations, monotonicity-comparison pairs, and policy-pair rankings. They should not be interpreted as respondent-level sampling uncertainty from re-estimating the full 6-parameter conditional-logit frontier unless explicitly stated.

J.1 Bootstrap Procedure

To quantify uncertainty, we compute bootstrap confidence intervals tailored to the specific unit of analysis for each evaluation metric.

For **held-out prediction metrics** (Figure 1 of the main manuscript reports the primary exact-task multinomial validation; respondent-level paired cluster bootstrap for the paired log loss

differences is reported below), the DCE data possess a nested panel structure where multiple choice observations are clustered within individual respondents. To strictly account for within-respondent correlation and avoid underestimating standard errors, we employ a **respondent-level cluster bootstrap**. Specifically, we resample the 205 unique respondents (whose choices constitute the 1,230 task-level multinomial observations) with replacement, maintaining the intact within-respondent choice structure. We perform 2,000 replications to ensure stable interval estimation.

For **frontier-deviation metrics** (Table S13), the evaluation operates on the fixed 64-state policy grid rather than respondent-level choices. Therefore, we compute frontier-deviation intervals by resampling the 64 policy-grid states with replacement using 2,000 replications. For **monotonicity violation rates** (MVR), we resample comparable attribute pairs. For **policy-ranking consistency** (Table S11 in this supplement, discussed in the main text Policy Ranking Consistency section), we report no confidence intervals and no significance test, because the 15 unordered policy pairs share six underlying policy effects and are not 15 independent Bernoulli trials; the metric is reported as a descriptive share only.

These intervals provide uncertainty summaries for the finite evaluation objects used in the paper: policy-grid states, held-out matched respondents, monotonicity-comparison pairs, and policy-pair rankings. They are not a substitute for external validation against realized uptake or for respondent-level uncertainty from re-estimating the full 6-parameter conditional-logit DCE frontier. The paired bootstrap intervals condition on the fitted training-split DCE model and the realized averaged LLM task probabilities; they quantify held-out respondent sampling uncertainty and do not incorporate DCE-estimation or LLM-generation uncertainty.

J.2 Held-Out Prediction Uncertainty

Table S12 reports respondent-level paired cluster bootstrap confidence intervals for the primary exact-task multinomial validation. The paired differences are:

- $\Delta LL_{\text{EFR-Raw}} = 0.9233 - 1.0200 = -0.0967$ (EFR improves over Raw LLM)
- $\Delta LL_{\text{EFR-Pure}} = 0.9233 - 0.9112 = +0.0121$ (EFR within 1.3% of Pure-DCE)

The bootstrap resamples 205 respondents with replacement (2,000 replications), maintaining the within-respondent cluster structure (6 choices per respondent). The 95% confidence intervals quantify the uncertainty in these paired differences.

Table S12: Respondent-level paired cluster bootstrap for exact-task multinomial log loss ($N = 205$ respondents, 1,230 choices; 2,000 replications).

Comparison	Point Estimate	95% CI
EFR vs. Raw LLM ($\Delta LL_{\text{EFR-Raw}}$)	-0.0967	[-0.1468, -0.0469]
EFR vs. Pure-DCE ($\Delta LL_{\text{EFR-Pure}}$)	+0.0121	[-0.0234, +0.0472]

J.3 Frontier-Alignment Uncertainty

Table S13 reports bootstrap intervals for the primary frontier-alignment diagnostics discussed in the main text. The intervals support the main Qwen2.5-72B-Instruct alignment result: the MSE and Spearman ρ intervals do not overlap between EFR and the unconstrained LLM. The MVR-Wait point estimates for Qwen2.5-72B-Instruct also do not overlap (raw 0.6250, EFR 0.3021); the bootstrap intervals, however, are wide and overlap on the 64-state grid (state-level resampling leaves

the 16 (eff, se) cell pattern close to identical across replicates), so the MVR-Wait improvement is best read from the point estimates and the cross-model ranking rather than from a within-Qwen significance test. The side-effect monotonicity diagnostic is weaker: for MiroThinker-1.7-mini, EFR and the unconstrained LLM produce identical MVR-SE point estimates (0.3333) and identical bootstrap CIs [0.240, 0.427], indicating that the EFR anchor at $\lambda = 0.25$ does not transmit a sufficient side-effect signal on this frontier.

K Covariate Balance Check: Matched vs. Unmatched Subset

To assess the representativeness of the 22.1% matched held-out subset ($N = 813$ *alternative-level binary-outcome* rows from $N = 205$ held-out respondents (each row is one of three alternatives within one of six tasks; the held-out pool contains $N = 3,690$ alt-rows; after excluding one anomalous wait=2 alt-row, 3,689 rows remain under the intended design, of which 813 are matched and 2,876 are unmatched)) used in Table 1 of the main text and in Table S7 of this supplement, we compare its covariate distribution against the unmatched held-out alt-row subset ($N = 2,876$). Table S16 reports the means and Standardized Mean Differences (SMD) for key DCE attributes and the choice rate.

The results confirm that the matched subset systematically differs from the unmatched subset on all observed dimensions (all SMDs > 0.1). The root cause is structural rather than sampling-based: the 64-state LLM grid evaluates only specific integer attribute levels, inherently oversampling longer waits, higher efficacy, and higher side-effect burdens. Consequently, the matched subset serves as a harder, non-representative test bed for the specific attribute profiles covered by the grid. Importantly, because all 205 respondents are present in both subsets, this covariate imbalance does not compromise the respondent-level cluster bootstrap intervals reported in the main text.

L Replication Materials

Replication materials are available in a public GitHub repository: <https://github.com/REDACTED/behavioral-digital-twins>. The repository provides the questionnaire, code, de-identified aggregate/grid-level inputs, model outputs, and derived analysis datasets required to reproduce the reported LLM and frontier-alignment analyses. Individual-level DCE responses are not publicly released because of participant confidentiality.

M Out-of-Design Narrative Evaluation (Supplementary Table Reference)

The main text (Section 4.3, Table 2 of the main text) reports the out-of-design Panel A and Panel B narrative-policy evaluation on the 20 paired scenarios (40 narrative descriptions) with MVR-Wait on the 8 wait-varying pairs (Panel A: 6; Panel B: 2), MAD/ ρ on the 12 narrative-only pairs, and the two-model comparison (Qwen2.5-72B-Instruct and DeepSeek V4 Pro; MiroThinker-1.7-mini was not administered on the OOD panel because its 64-state grid evaluation was used to validate the EFR anchor as a standalone test, not to run the OOD narrative scenarios). The four-panel diagnostic figure (MVR-Wait / Spearman ρ / subgroup heterogeneity / PRC) that combined the 20 paired OOD scenarios with the 813-row matched-subset subgroup analysis is omitted from this supplement because the rank-correlation panel falls inside the construction-only null band; the table-based diagnostics (Table 2 of the main text) are the authoritative source for the OOD

narrative results. The out-of-design evaluation is restricted to the two-model Qwen2.5-72B-Instruct and DeepSeek V4 Pro panel; the third model (MiroThinker-1.7-mini) is reported only on the 64-state grid in the main text.

N Primary Exact-Task Multinomial Validation: Six Canonical DCE Tasks

This appendix reports the primary exact-task multinomial held-out validation on the six canonical DCE tasks (Choicesets 1–6) administered to all 1,027 respondents. The validation uses a local replication run with Qwen2.5-72B-Instruct via Ollama (temperature 0.7, 10 repetitions per task) to generate LLM predictions for direct comparison with the empirical choice data.

N.1 Design and Setup

All 1,027 respondents completed the same six choice tasks, each presenting two non-opt-out alternatives (A, B) and an opt-out alternative (C). The six administered tasks comprised 12 unique non-opt-out profiles, covering 12 of 216 candidate profile cells (5.6%). A post-hoc dominance audit of the six A-versus-B profile pairs confirmed no strict dominance across the five monotonic attributes; two tasks (Tasks 1 and 3) exhibited near-dominance with one alternative winning on four of five attributes, but each retained at least one attribute favoring the other alternative. We hold out 20% of respondents ($N = 205$) using a seeded random split (seed = 2026; 822 training, 205 test) and use their choices on the six canonical tasks for validation. Exact log loss values are reported in Table S16.

The LLM validation uses respondent-facing multinomial prompts that elicit explicit probabilities $P(A)$, $P(B)$, and $P(C)$ for each task. Three methods are compared:

- **Pure-DCE:** Refit 6-parameter conditional logit on the 80% training split, multinomial softmax over $\{A, B, C\}$.
- **Raw LLM:** Direct LLM $P(A)$, $P(B)$, $P(C)$ averaged over 10 repetitions per task.
- **EFR** ($\lambda = 0.25$): Convex blend in the three-alternative probability simplex. For each task t , the DCE training-split model yields $\mathbf{p}_t^{\text{DCE}} = \text{softmax}(V_{At}, V_{Bt}, V_{Ct})$ where $V_{Ct} = \hat{\beta}_{\text{ASC}}$ (the opt-out alternative-specific constant), and the LLM yields $\mathbf{p}_t^{\text{LLM}} = [P(A), P(B), P(C)]$. The EFR vector is:

$$\mathbf{p}_t^{\text{EFR}} = \lambda \mathbf{p}_t^{\text{LLM}} + (1 - \lambda) \mathbf{p}_t^{\text{DCE}}$$

with $\lambda = 0.25$. Because both $\mathbf{p}_t^{\text{LLM}}$ and $\mathbf{p}_t^{\text{DCE}}$ lie on the probability simplex ($\sum_j p_j = 1$, $p_j \geq 0$), the convex combination $\mathbf{p}_t^{\text{EFR}}$ also lies on the simplex and $\sum_j p_{t,j}^{\text{EFR}} = 1$ by construction.

Extension from binary formulation. The main-text EFR equation $\pi_{\text{EFR}}(y = 1 \mid \mathbf{x}) = \lambda \pi_{\text{LLM}}(y = 1 \mid \mathbf{x}) + (1 - \lambda) P_{\text{emp}}(y = 1 \mid \mathbf{x})$ is a binary policy-versus-opt-out formulation where P_{emp} is the sigmoid of the utility gap. For the exact-task categorical validation, EFR is extended from this binary formulation to the full three-alternative probability simplex by convexly combining the LLM and training-split DCE probability vectors as defined above. The binary and vector formulations are equivalent under IIA when the choice set contains only one non-opt-out alternative and the opt-out; with two non-opt-out alternatives (A, B) plus opt-out (C), the softmax vector is the natural generalization.

Held-out leakage prevention. All DCE components used in the held-out comparison, including the empirical component of EFR ($P_{\text{emp,train}}$), were estimated exclusively from the 822 training respondents. The 205 held-out respondents were not used in any coefficient estimation for the validation metrics reported in Table S16.

N.2 Results: Six-Task Multinomial Log Loss

N.3 Interpretation

The results show a clear ranking: Pure-DCE outperforms all other methods, but EFR substantially improves over Raw LLM and approaches the Pure-DCE performance:

- **EFR vs. Raw LLM:** 9.5% reduction in log loss (from 1.020 to 0.923), demonstrating material improvement in exact-task predictive performance.
- **EFR vs. Pure-DCE:** Within 1.3% of the empirical frontier (0.923 vs. 0.911), showing that EFR recovers most of the DCE model’s predictive performance while retaining a non-zero LLM component.
- **All models vs. Uniform:** All three model-based approaches outperform the equal-probability benchmark ($\log 3 \approx 1.099$), indicating better-than-uniform prediction on the six administered DCE tasks.

Scientific conclusion: EFR does not outperform the DCE model within the original structured choice domain, but it substantially improves on the unconstrained LLM and recovers most of the DCE model’s predictive performance while preserving an LLM component for narrative extension.

N.4 Model and Inference Condition

Qwen2.5-72B-Instruct local implementation. This validation uses a local replication with Qwen2.5-72B via Ollama (model: `qwen2.5:72b`, quantization: `Q4_K_M`, Ollama version 0.3.0). Inference parameters: temperature 0.7, top-p 1.0, max tokens 512, 10 repetitions per task. The inference condition differs from the original Hugging Face endpoint (Table S2) in quantization and sampling implementation; exact numerical correspondence is not guaranteed. Endpoint and model identifiers for DeepSeek V4 Pro and MiroThinker-1.7-mini are reported in Table S2.

Prompt template (exact-task multinomial validation). The respondent-facing multinomial prompt for the six canonical DCE tasks is:

You are evaluating a vaccination decision. The respondent is deciding whether to accept the proposed COVID-19 vaccine. Read the policy description below and decide.

Alternative A

Waiting time: [VALUE] months

Vaccine effectiveness: [VALUE]%

Risk of side effects: [VALUE]%

Cash incentive: [VALUE] RMB

Vaccine origin: [Domestic/Imported]

Alternative B

Waiting time: [VALUE] months

Vaccine effectiveness: [VALUE]%

Risk of side effects: [VALUE]%

Cash incentive: [VALUE] RMB

Vaccine origin: [Domestic/Imported]

Alternative C

Do not receive the vaccine / Opt out

What is the probability that the respondent chooses each alternative?

Report $P(A)$, $P(B)$, and $P(C)$ as probabilities between 0 and 1 that sum to 1.

Format: $P(A)$ =[value], $P(B)$ =[value], $P(C)$ =[value]

Parser and normalization rules.

1. **Extraction:** Parse $P(A)$, $P(B)$, $P(C)$ using regex pattern $P((A|B|C)s*=s*([0-9.]+).)$
2. **Validation:** Check that all three probabilities are present and parseable. If any missing, mark as `PARSE_FAIL`.
3. **Sum check:** If $|P(A)+P(B)+P(C)-1.0| > 0.01$, apply proportional normalization: $P'(k) = P(k)/\sum_j P(j)$. In the analyzed outputs, no normalization was required (all probabilities summed to 1.0 within floating-point precision).
4. **Clipping:** Final probabilities clipped to $[0.001, 0.999]$ to avoid $\log(0)$.
5. **Position bias check:** Alternatives A/B presentation order matches the original DCE questionnaire. No counterbalancing was performed for the primary validation because the empirical DCE frontier is order-invariant. To assess potential LLM-specific position bias, we conducted a sensitivity experiment swapping A/B labels on the six canonical tasks (60 additional generations). The Raw LLM log loss changed from 1.0200 (original order) to 1.0307 (swapped order), a relative difference of 1.05%, suggesting minimal position bias for the evaluated models.

Missing output handling. If the LLM fails to generate a parseable response (e.g., refusal, malformed output), the response is marked as `PARSE_FAIL` and excluded from the mean calculation. In practice, parse failures were rare ($< 1\%$ for Qwen2.5-72B).

Table S13: Bootstrap intervals for selected frontier-alignment diagnostics (2000 state-level bootstrap replicates with replacement; MVR-Wait reported as point estimate because the 16 (eff,se) cell-level structure makes the MVR a step function on the grid).

Model	Method	Metric	Estimate	
Qwen2.5-72B-Instruct	Unconstrained LLM	MSE	0.0258	[0.0174, 0.0350]
Qwen2.5-72B-Instruct	EFR Anchor	MSE	0.0016	[0.0011, 0.0022]
Qwen2.5-72B-Instruct	Unconstrained LLM	MVR-Wait	0.6250	point
Qwen2.5-72B-Instruct	EFR Anchor	MVR-Wait	0.3021	point
Qwen2.5-72B-Instruct	Unconstrained LLM	Spearman ρ	0.0608	[−0.1784, 0.3042]
Qwen2.5-72B-Instruct	EFR Anchor	Spearman ρ	0.6886	[0.5266, 0.8012]
DeepSeek V4 Pro	Unconstrained LLM	MSE	0.0379	[0.0266, 0.0505]
DeepSeek V4 Pro	EFR Anchor	MSE	0.0024	[0.0017, 0.0031]
DeepSeek V4 Pro	Unconstrained LLM	MVR-Wait	0.3438	point
DeepSeek V4 Pro	EFR Anchor	MVR-Wait	0.0729	point
DeepSeek V4 Pro	Unconstrained LLM	Spearman ρ	0.3025	[0.0714, 0.5123]
DeepSeek V4 Pro	EFR Anchor	Spearman ρ	0.8123	[0.6832, 0.8885]
MiroThinker-1.7-mini	Unconstrained LLM	MSE	0.0872	[0.0662, 0.1107]
MiroThinker-1.7-mini	EFR Anchor	MSE	0.0054	[0.0041, 0.0070]
MiroThinker-1.7-mini	Unconstrained LLM	MVR-Wait	0.1042	point
MiroThinker-1.7-mini	EFR Anchor	MVR-Wait	0.0833	point
MiroThinker-1.7-mini	Unconstrained LLM	Spearman ρ	0.5306	[0.3070, 0.6966]
MiroThinker-1.7-mini	EFR Anchor	Spearman ρ	0.7564	[0.5849, 0.8860]

Note. Unlike the held-out prediction metrics in Table S7 (held-out prediction metrics), which employ respondent-level cluster bootstrapping to account for the nested panel structure of DCE choices, the frontier-alignment metrics here are evaluated on the fixed 64-state policy grid. Therefore, frontier-deviation intervals are computed by resampling the 64 policy-grid states with replacement (2,000 replications). MVR-Wait is reported as a point estimate: the 16 (eff,se) cell-level structure makes the MVR a step function on the grid, so a state-level bootstrap yields degenerate intervals; a comparable-attribute-pair bootstrap (resampling (eff,se) cells) gives the same point estimate by construction because the frontier-deviation ranking is invariant under cell-level resampling. The MiroThinker-1.7-mini side-effect diagnostic (MVR-SE = 0.3333 for both unconstrained and EFR) is omitted from this table: the corrected 6-parameter specification (Section 3 of the main text) makes the side-effect coefficient the smallest empirical signal among the five attributes, and the EFR anchor at $\lambda = 0.25$ does not transmit a sufficient side-effect correction to move MiroThinker’s MVR-SE off the unconstrained value; the failure mode is documented in Section 4.3 of the main text (OOD panel-level diagnostics note).

Table S14: Attribute-level mapping between the administered DCE design and the 64-state LLM evaluation grid. The matched subset is restricted to cells where the DCE design level and the grid level coincide as a `float` value (no rescaling, no interpolation).

Attribute	DCE Levels	Grid Levels	Literal Intersection (matched subset)
Wait Time (months)	{0, 1, 3, 6}	{0, 2, 4, 6}	{0, 6}
Vaccine Efficacy	{0, 0.5, 0.7, 0.95}	{0.3, 0.5, 0.7, 0.9}	{0.5, 0.7}
Side-Effect Burden (coded level)	{0, 1, 2, 3}	{0, 1, 2, 3}	{1, 2, 3}
Cash Incentives (RMB)	{0, 50, 200, 800}	0 (held)	— (not matched)
Vaccine Origin	{0, 1}	0 (held)	— (not matched)

Table S15: Covariate balance between matched and unmatched held-out subsets.

Variable	Matched Mean	Unmatched Mean	SMD
Wait Time (months)	2.77	1.33	0.58
Vaccine Efficacy	0.59	0.45	0.49
Side Effects	1.95	1.16	0.79
Cash Incentives (RMB)	325	207	0.38
Vaccine Origin (0=dom, 1=imp)	0.50	0.28	0.45
Choice Rate ($y = 1$)	0.528	0.278	0.53

Note. The matched subset consists of held-out observations whose $(wait, eff, se)$ cell is in the literal overlap between the intended DCE design and the LLM grid on the three focal attributes (waiting time, efficacy, and side-effect burden). SMD denotes the Standardized Mean Difference. The DCE design administers $(wait, eff, se) \in \{0, 1, 3, 6\} \times \{0, 0.5, 0.7, 0.95\} \times \{0, 1, 2, 3\}$ (the four real wait design levels; the raw coded data contain one additional 1-row outlier at $wait = 2$ months from a single respondent which is excluded as a data anomaly because it is not an administered design level); the 64-state grid uses $(wait, eff, se) \in \{0, 2, 4, 6\} \times \{0.3, 0.5, 0.7, 0.9\} \times \{0, 1, 2, 3\}$. The literal overlap on the three focal attributes (cash incentives and vaccine origin are held at grid reference values and do not participate in the matching criterion) is $wait \in \{0, 6\}$, $eff \in \{0.5, 0.7\}$, $se \in \{1, 2, 3\}$, giving $2 \times 2 \times 3 = 12$ candidate cells of which 11 are populated in the matched subset (the $(wait = 0, eff = 0.7, se = 2)$ cell is not observed in the matched non-optout pool). Cash incentives (DCE levels {0, 50, 200, 800}, held at grid reference value 0) and vaccine origin (DCE levels {0, 1}, held at grid reference value 0) are not part of the matching criterion; DCE cells with $cash > 0$ or $origin = 1$ are matched on $(wait, eff, se)$ alone. The matched subset ($N = 813$ alternative-level rows from 78 task-level 2-alternative non-opt-out tasks) is therefore not a random sample of the unmatched subset ($N = 2,876$ alt-rows under intended design); the SMDs in the table above quantify the systematic differences. Crucially, all 205 held-out respondents contribute to both subsets; thus, the imbalance is strictly at the observation (attribute-level) rather than the respondent level.

Table S16: Primary exact-task multinomial held-out log loss on six canonical DCE tasks ($N = 205$ respondents, 1,230 choices). Lower is better. The conditional logit is refit on the 80% training split. LLM predictions generated via local Ollama Qwen2.5-72B (temperature 0.7, 10 repetitions per task).

Method	Log Loss	Note
Pure-DCE	0.911	Refit 6-param conditional logit; best performance
EFR ($\lambda = 0.25$)	0.923	Convex blend; approaches Pure-DCE within 1.3%
Raw LLM	1.020	Direct multinomial LLM prediction
Uniform (1/3 each)	1.099	Equal-probability baseline

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